LENDÜLET 2024

KUTATÁSI TERV / PROJECT PROPOSAL

A pályázó kutatócsoport-vezető neve / Name of the prospective Principal Investigator (PI) Lengyel Balázs / Balázs Lengyel
A pályázat címe / Application Title Innováció és egyenlőtlenség dinamikus városi hálózatokban / Dynamic urban networks of innovation and inequality
Lendület kategória / Lendület category H
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I. The prospective PI's narrative CV

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After being a public servant, I started to work at the Hungarian Academy of Science (HAS) in 2006 at the Centre for Regional Studies. In 2008, I moved to the Institute for World Economics and defended my PhD thesis in economics at the Budapest University of Technology and Economics in 2010. After a career break spent with an Indonesian state scholarship in 2010-2011, I started to work on social networks at the International Business School Budapest. In 2013, I joined the Institute of Economics from where I visited Umea University in 2014-2015. A major career milestone was my research visit at MIT in 2016-2017. During that period, I was awarded the Lendület scholarship that allowed me to establish the Agglomeration and Social Networks Lendület Research Group (ANET Lab) in 2017 at the Institute of Economics where I work there as Senior Research Fellow since 2018. In 2020, I was invited to establish the Laboratory for Networks, Innovation and Technology (NETI Lab) at the Corvinus University of Budapest (CUB). In 2023, I became an Associate Professor at CUB.

Research achievements
In the past years, I have successfully introduced quantitative techniques of network science to the research on technological and economic advance of regions and cities covering a large variety of topics. I have published 61 journal articles, out of which 46 came out in international journals. I have received 2130 citations to my work (google scholar, H index is 27).

Being an active contributor to the economic geography discourse, I introduced large-scale co-worker network analysis to understand labor market outcomes such as firm- and regional productivity growth using administrative data on matched employer-employee records in Sweden and in Hungary. My related papers have been published in Journal of Economic Geography, Economic Geography, and The Journal of Technology Transfer. I have also investigated the role of collaboration networks among inventors to better understand how radical innovation is created in regions, how regions diversify into new technologies and how persistent collaboration describes innovation systems of European regions. Related papers are published in Research Policy, Economic Geography, European Planning Studies and PLoS ONE. Further work on the role of foreign-owned companies, automation in regional economic development has been published in several papers in Regional Studies, Papers in Regional Science, among other journals.

I am also doing research in the interdisciplinary field of complex social networks, analyzing their role in creative success and inequalities within and across cities. With my co-authors, we were invited to publish in a Nature supplement in 2020 to discuss how scientific collaboration across global research hubs is changing. My work on urban mobility networks and segregation in cities have been published in Scientific Reports, Environment and Planning B, and in EPJ Data Science. Besides all these achievements, I am very proud of my decade-long work with iWiW, a Hungarian online social network, that has helped us to model the spatial diffusion of innovation, the social network determinants of mental health, and the relationship between network structure and income inequality dynamics within cities. These contributions were showcased in 2022 in a national exhibition in Budapest that collected the top achievements of Hungarian science from the last 200 years (links to the exhibition, and to our exhibited animation). Related work has been published in Nature Communications, Scientific Reports, PLoS ONE, Network Science, among other outlets.

Peer recognition

I was a recipient of the Lendület (2017), Bolyai (2017), Eötvös (2016), and Rosztoczy (2015) awards. Being the PI of five Hungarian research grants (total sum around 700K EUR), each of my projects was evaluated as outstanding. In 2022, my ERC Consolidator grant proposal was selected for interview, but I did not receive funding.

The number of invitations to international events is increasing. I was a lecturer at the Oxford Summer School on Economic Networks in 2020 and in 2023, the POLISS International PhD course in 2022, and the Winter School on the Geography of Innovation in 2024. I have been invited to deliver talks at the MIT Media Lab, the Harvard University, Northeastern University, University of Manchester, University of Bristol, University of Bordeaux, the Tinbergen Institute, Central European University, Complexity Science Hub Vienna, Toulouse University, and the Joint Research Centre of the European Commission, among other global leading institutions. I am a keynote speaker at the Global Conference of the Regional Science Association International in 2024.

I was elected to the vice-president of the Regional Science Committee of the Hungarian Academy of Sciences in 2020 and have been elected multiple times to the Board of the Hungarian Regional Science Association.

Other contributions to the research community

Being the head of ANET Lab at the Hungarian Research Network (previously Hungarian Academy of Sciences) from 2017 and the co-director of NETI Lab at Corvinus University of Budapest from 2020, I have been directing four postdocs, four PhD students, and ten research assistants. The alumni of my team have positions at University of Amsterdam, Northeastern University London, Complexity Science Hub Vienna, and Umea University.

At the Hungarian Research Network, I have been coordinating a new initiative that aims to build data processing capacities for the research institute. A new group was established under my guidance and has become a very successful project. At the Corvinus University of Budapest, I participated in the Scientific Research Committee under the leadership of the Vice Rector for research. I was a main contributor to the "Collective Learning Data Lab" ERA-Chair grant application that awarded 2.5 million Euros to Prof. César Hidalgo, to establish his new laboratory at the Corvinus University of Budapest.

As elected representative of the Hungarian regional science community, I have been active in organizing scientific events including seminar series and conference special sessions, in the coordination of the best paper award ceremony, among further day-to-day management tasks. I have been Committee member of PhD defences at Birmingham University, the Central European University, University of Pécs, University of Győr.

In 2018, I was invited to join the Hungarian Young Academy (HYA) as a founding member. With HYA, we have consulted with scientific funds about general practices of refereeing processes and could achieve more inclusive research calls. My main activity in HYA has been the coordination of its' main research project that mapped the work conditions of early career researchers in Hungary (link to report) and is currently used as the basis for science policy recommendations of HYA.

My service for the international scientific community embodies in conference organization and refereeing. I participated in the organization of the IC2S2 2023 conference as the member of the program chair committee, and at the ERSA 2022 congress as a local organizer. I am regularly contributing as a program or scientific committee member for the NetSci, NetSciX, Complex Networks, and the Geography of Innovation conferences. I served as a referee for 32 journals. I am the main organizer of the next Geography of Innovation Conference that will take place in 2026 in Budapest. I serve as editorial board member at the Hungarian Statistical Review and the Geografiska Annaler B.

II. Short description of the research project planned for five years

Increasing inequalities together with the rising concentration of global population in metropolitan areas are challenging the way we think about cities [1,2]. On the one hand, cities provide efficiency gains for firms and individuals. A principle frequently referred to as agglomeration economies claims that densely populated areas prosper due to efficient sharing of public goods, matching on markets, and learning [3–5]. Cities are home of diverse knowledge, mixing of which made cities the engines of scientific discovery and technological innovation [6–10] and put them to the front of tackling humanity's global challenges [11]. On the other hand, cities are known for high levels of segregation [12,13]. The contrast across rich and poor neighborhoods, urban poverty, and crime signal that people have unequal access to the advantages of cities around the globe [14]. Inequality is a top social policy priority, it fosters political populism and radicalization, and expose society to economic transformation, health or climate crises [15–18].

Innovative cities tend to have highly unequal wealth distributions [19–21] and many seek to understand how urban policy and planning can help innovation [22] and solve the problem of exclusion [23–25,115]. Yet, a growing community calls for research into how relational micro-processes are linked to city level outcomes [26–28] because our knowledge in this matter is still very limited.

Social networks have been useful tools to characterize this dual facet of urbanization. Interpersonal relations facilitate information sharing more than mere co-location [29]. Hence, the structure of social networks qualify agglomeration economies because it influences how fast and far information can travel and how new knowledge is created [5,30]. Accordingly, economic progress and technological innovation stand out in cities where social networks are dense because sharing, matching, and learning can be more efficient [31,32] and because the combination of new ideas is easier [33,34]. However, social networks are not favoring the urban population equally and can induce inequalities through the mechanisms of segregation [35–37]. Resources that can be mobilized through connections, often referred to as social capital [31,38], depend on the position in the network so that central individuals benefit more than peripheral ones [39–41].

Spatial research, that this proposal aims to contribute to, has mainly focused on the structure and the dynamic changes of networks (i.e. the creation, persistence, and dissolution of ties) [34,42–47]. A central approach builds on the notions of network closure – the tendency that agents with a common peer are likely to form a tie – that facilitates cohesion and trust among similar peers [36,48,49] and bridging – the establishment of links across loosely connected parts of networks – that provides access to diversity [39,50,51]. These two processes lead to fragmented networks, a general structure found in many types of networks in nature and society [52,53]. In fragmented networks, cohesive communities are loosely connected to one another, having direct consequences on innovation and inequality. On the one hand, fragmented networks are thought to combine advantages of closure and bridging that favors innovation because such networks provide access to diverse knowledge through bridges and an efficient learning environment in cohesive communities [39,50,51,54]. On the other hand, fragmented networks can increase inequalities as well in case communities of the rich and the poor are separated because the poor do not have access to social capital of the rich [36,55]. Despite the above insights, the structural approach lacks a comprehensive view on the role of social networks in cities by missing crucial elements: content and dynamics of information flows. Although social networks are theorized to have an impact by transmitting information, we know almost nothing about the contextual, spatial, and dynamic nature of these flows.

This research niche is a major constraint for understanding innovation and inequality in cities since diffusion of advice or new knowledge is key for learning and innovation [56,57], while inefficient flows of novel or job-related information are sources of rising inequalities in innovation adoption and on the labor market [36]. Communication is temporal, differs by content, by spatial context, and parties involved [41,58,59] but these features are still missing from the scientific discussion. Overlooking the content and dynamics of social interactions, however, limits insights on network effects by hiding what dimensions of separation and mixing matter, how bridging facilitates diffusion and how it influences knowledge creation and unequal progress.

The problem has a strong spatial character. Although online platforms are major channels of communication, social interactions still concentrate around home and work [60–62]. Commuting and mobility to third locations enable face-to-face meetings across the city that is crucial for learning, and new knowledge creation [63] and have been found to foster social mixing across socio-economic strata as well [64]. Yet, how urban mobility facilitate bridging and across which groups, is still unknown. The problem is, however, very timely, since mobility restrictions and the wide adoption of home-office have reduced the share of bridging ties and intensifying network closure during COVID-19 lockdowns [65]. These trends might have a consequence on innovation by limiting the circulation of diverse knowledge [65] and can increase inequalities by limiting access to social capital [66].

In this project, we will answer the research question: **How do content and dynamics of information flows in spatial social interactions impact knowledge creation and inequality in cities?**

To answer the question, the project will combine approaches of urban mobility and the network diffusion to achieve groundbreaking empirical advances in causal inference and models on dynamic network processes. These breakthroughs will enable us to advance theory on network divergence and convergence of urban innovation and inequalities.

In the following, we review the main areas of our contributions to the state-of-the-art literature and identify the breakthroughs offered in the project. Next, we introduce the details of the empirical work packages including risk management in each work package, the resources used, and the timeline.

Main areas of contributions and breakthroughs

We will target three interconnected fields: (1) the foundations of divergent and convergent network dynamics, (2) the role of places and urban mobility in network bridging, (3) the diffusion of new information in spatial social networks. These are the main blocks of advancing the state-of-the-art in the research project.

Area 1. Network divergence and convergence. Individuals tend to connect with similar peers – referred to as homophily or assortativity – [49,114] and are forming communities that close triangles across peers [48]. Network closure can lead to divergence in case communities that differ in knowledge specialization or income are not or only loosely connected. However, social networks are small worlds because even few bridging ties can make network access short and induce mixing across diverse communities [67].

For example, majority of collaborative projects in science and technology focus on specific knowledge domains around which inventor and academic communities specialize. However, access to new knowledge and potential for new combinations can motivate connection across differently specialized groups [33,68–70] (Figure 1) and across locations [71,72]. This bridging can result in creation of radical novelty, high impact and interdisciplinary science [73,74] that is also demonstrated in our own research [75]. Although such mixing is important for the long-term development of cities [42,43], how this process happens in and across cities and influence knowledge production in urban areas, is not clear.

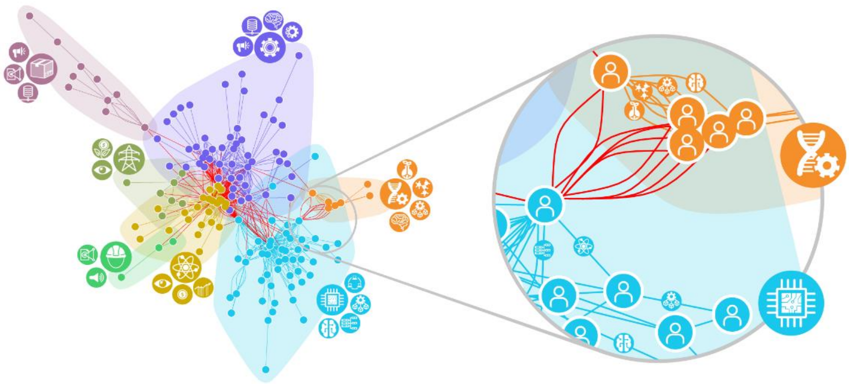

***Figure 1.** Network structure of innovative collaborations* [75]*. Collaborations of differently specialized co-inventor communities result in more radical novelty. The type of knowledge flows is not observed.*

The state-of-the-art approach of describing urban innovation with spatial collaborations is primarily structural [34,76]. Investigations of network processes in this literature are mostly restricted to the creation and maintenance of ties, while underlining micro-mechanisms and processes remain unobserved [42,43,47,75,77,78]. This leads to oversimplified explanations of network evolution and knowledge creation missing the type of knowledge shared among peers. Yet, knowledge diffusion is far from being homogenous. For instance, complex knowledge concentrates in cities [79] because it is difficult to transfer across distance and requires frequent connections to share [80]. Hence, we need to distinguish the type of knowledge shared in networks to uncover how closure and bridging ties influence knowledge creation in cities.

We know even less about divergent network dynamics of urban wealth. Social networks maintain and even amplify inequalities when economic status plays a role in social tie formation [35,36]. Segregated networks have been found to induce inequalities in towns where income disparities are high (Figure 2), suggesting that separated networks of the rich and the poor lead to wealth divergence [55]. Yet, it is not clear whether and how new links between rich and poor communities can induce catching up effects. Such links might provide new social capital to the less advanced that can be used on the labor market, for entrepreneurship, or through changes in firm productivity. Yet, previous research on this matter is completely missing and the time and spatial dimensions of these processes demand a series of new explorations.

Breakthrough: Since complex communication and collaboration in cities have many layers, the main challenge is disentangling the types of closure and bridging. Recently available data including social media, and software version-control systems, and new methods to deal with the large amount of information in scientific publications allows for observing the content of closing and bridging interactions. This can be used not only to classify connections and separate recent and old relations of strata, partnership, or acquaintance in longitudinal and large social networks, but also to investigate the temporal patterns of this dynamic. Advancing the state of the art by including the content of interactions allows for breakthrough theories regarding network divergence and convergence and their impact on urban innovation and inequality progressing two methodological fronts: the role of space and diffusion of new information, which will follow next.

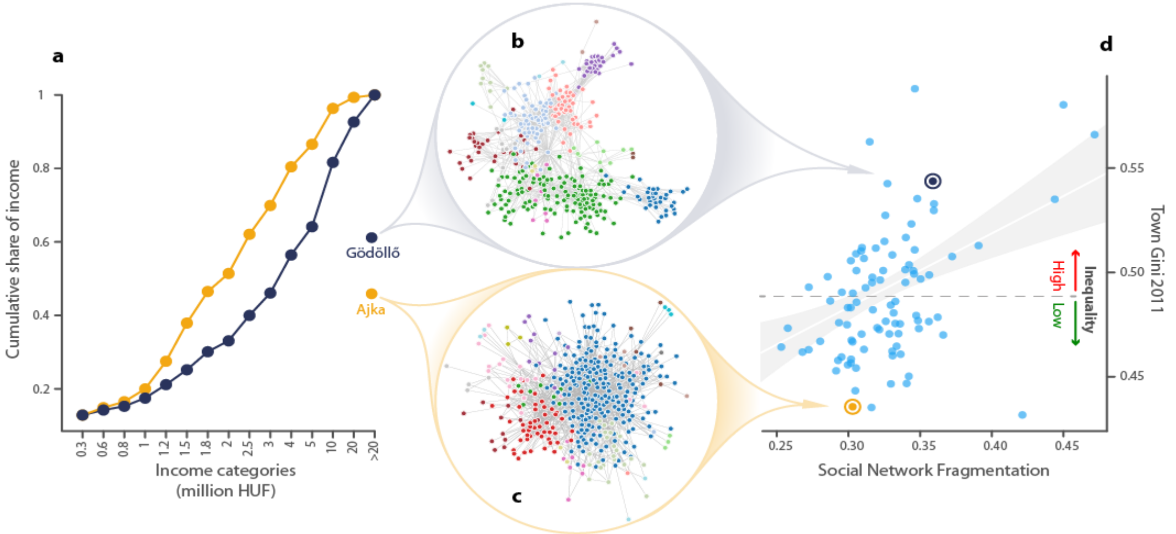

Figure 2. *Network structure and income inequality in towns* [55]*. **a)** Income distribution of an equal (Ajka) and unequal (Gödöllő) town in Hungary. **b-c)** The unequal town has a more fragmented network than the equal town. **d)** Network fragmentation and income inequality is correlated across towns.*

Area 2. The role of physical space and places in social interaction. Geographical locations are predictors of economic prospects [81] and innovation in cities [82,83] signaling contrasts within areas; while distance is an impediment of social interaction [84]. Spatially separated local communities in innovative collaboration [44,85] and other types of social networks [60,84,86] within and across cities alike limit the available diversity for knowledge creation and leads to income divergence across segregated neighborhoods. On the contrary, connections to distant and prosperous places can help knowledge creation and widen the pool of resources for individuals [32,41] but the breadth and depth of these process are largely hidden.

As an enabler of social interaction, geographical places provide opportunity for face-to-face interactions that can result in lasting relations [63,87,88]. Thus, mobile scientists and inventors can link cities transferring knowledge and building important collaborations for innovation [89]. Similarly, urban mobility and commuting across distant neighborhoods, that can now be analyzed with high resolution mobility, cellphone, and visitation data [26,90–92], can link citizens of various backgrounds [64]. The role of places in separation and mixing in the urban fabric has been in the focus of the planning literature for long [93,94]. Yet, it is still unclear how and what places can contribute to healthier social capital through facilitating lasting connections in cities [95].

The state-of-the-art discussion on the role of places in social interaction in cities concentrates around the question whether urban mobility and face-to-face meetings enabled by mixing in the urban fabric can mitigate social segregation. Many argue that most locations do not link individuals from various strata because spaces of social mixing are very difficult to enter and remain assortative instead [26,90,96]. Yet, our own research signals that urban mobility can generate network bridging across different neighborhoods and strata. Those individuals who commute to a distance that is above the city median, have less assortative social networks than those whose workplace and home are below the median commuting distance [64]. Further, social networks are significantly more fragmented in those cities where physical barriers hinder mobility, and income inequalities rise more intensively in these cities [55].

Breakthrough: The major difficulty for advancing this understanding is the identification of causality between mobility, social interaction, and outcomes in innovation and inequalities. Yet, with recently available data on mobility and interaction within cities and by using the exogenous variation in spatial and political segregation across places, one can identify the causal relation between networked interaction and outcomes in innovation and income. This opens the floor to better understand how improving access, across which places, and at what locations can create inclusive and innovative environments, across whom and on what timeframe.

Area 3. Diffusion of new information in networks. Social networks are thought to support economic and technological progress in cities by channeling the flow of new information [31–34,55,97]. However, surprisingly little is known on such flows in spatial social networks. Most related work focused on citations in patents or scientific publications [98–101] or used case studies [102] to investigate spatial flows. No doubt, information flows are much more complex than captured previously. Nevertheless, the mechanisms, and effects of spatial diffusion of new information are unexplored.

The state-of-the-art on information flow has been advanced in the network diffusion research. Novelty impacts collective outcomes through adoption that is driven by social influence embedded in networks: an individual is more likely to adopt novelty if his/her network neighbors have already adopted it. Yet, adoption can be limited in assortatively mixed societies because social influence creates diverging adoption dynamics [56,103]. For example, fragmented networks isolated by assortative mixing maintain inequalities because certain information diffuse in one network community but do spread to others [36]. This issue has many angles: (i) the origin of information matters in how extensive the spread can be [103,104]; (ii) certain information can break community boundaries while others cannot [59,105,106]; and (iii) the temporal nature of bridge opening and closure slows down diffusion across communities [107,108].

Despite the long traditions in geography [109] and recent calls for further research [56], except for [110,111], we know almost nothing about the social network mechanisms of spatial diffusion of innovation. The state-of-the-art in spatial network diffusion of innovation is limited to the spread of online services, due to access of sufficient data. This literature stresses that models of innovation diffusion on spatial social networks can predict adoption on the local level [110,111]. However, the problem of assortativity is still not solved [111], and geographical factors such as distance from the original location of the innovation as well as city size disturb the accuracy of model predictions [110].

Breakthrough: Assortative mixing [49] in networks and in cities makes social influence difficult to separate from susceptibility. This problem has been solved in field experiments, in which online communication links were blocked to measure how content of a message determines adoption [112]. Inspired by this literature, a breakthrough approach of spatial experiments can use mobility restrictions to examine changes of social interaction content and new information flows in firms, local communities, and across neighborhoods. Next, the generality of discoveries offered by this new approach can be enhanced in network diffusion models. Scientific publications provide data to detect novelty in science [75,113] and to trace the diffusion of publication abstract or keyword details across large networks of scientists. Fitting models on spatial collaboration networks can provide new insights into how different types of information flows across places, within and across network communities can impact knowledge creation and inequality in prestige.

III. Detailed work plan for the first three-year phase of the research project, with a breakdown of tasks by year, including envisaged output and publication activity

III.1 Work Packages

The project will answer the question and contribute to theory and methods in four empirical and one theoretical work packages (WPs), each of which will aim new insights on innovation and inequalities (Figure 3). Three of the WPs will target how dynamics of content in spatial social interaction within neighborhoods and companies (WP1), across neighborhoods (WP3), and across cities (WP4) relate to urban innovation and inequality. WP2 will include tests on alternative mechanisms of spatial social interaction to disentangle the impact of social network layers and workplaces.WP1 and WP3 will combine causal inference with diffusion of information. While WP2 will focus more on causality issues, WP4 will engage more with diffusion modeling. The WPs follow a vertical order and investigate innovation and inequalities on the micro, meso, and macro levels. Finally, we will summarize the empirical findings in a unified theoretical framework in WP5.

The strength of the empirical work is the rich combination of diverse and large datasets with innovative primary data collection across many cities. We will focus on Amsterdam, Budapest, London, and Stockholm, and the largest 15 metropolitan areas in the USA that represent cities of different size, economic development, and institutional background.

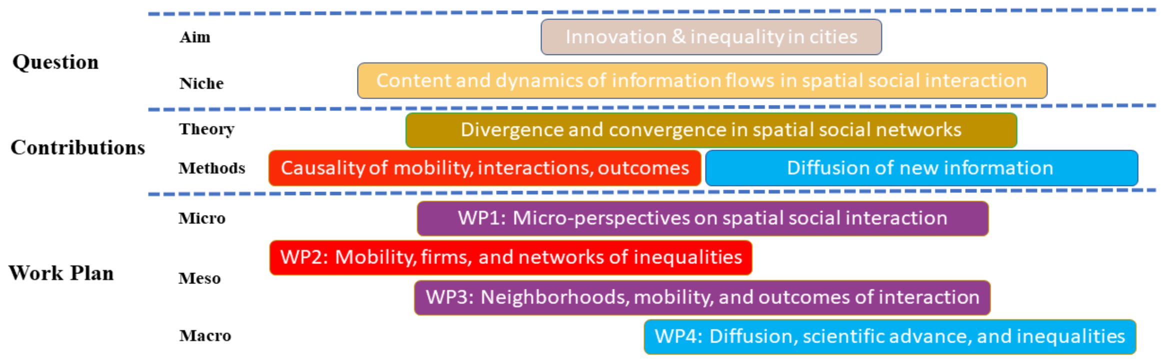

Figure 3. *Advancing the state of the art in the empirical work packages. WP 5 is not detailed in this plot, since it will cover the whole project.*

The work packages are broken down to projects. The brief research problems, data, methods, outputs, ethical principles, and risk management are presented by projects.

Work Package 1: Micro-perspectives on spatial social interaction

We aim detailed case studies to uncover micro-mechanisms of colocation, social mixing, and collaboration in firms by combining data donation techniques with survey methodology, and data from software version-control systems.

Project 1.1. Co-location, mixing, and collaboration in firms. Closure in collaboration networks provide efficient learning and the growth of productivity [54], while bridging connections provide access to diversity that can help the creation of novelty [39,51,68]. At the same time, assortative connections in terms of education and skills fragment collaboration networks at workplaces inducing inequalities within firms [143]. Despite co-location can facilitate the strength of links and can create new bridges, little is known about the role of co-location in catalyzing networked interactions of collaborative production. An exception is [65] that finds that collaboration networks became static and more siloed over the COVID lock-downs suggesting that co-location is needed for bridging and innovation at workplaces.

Data: In this project, we will analyze version control systems of companies' software development paired with innovative survey technologies [116]. Code documented and stored in git repositories can be used to construct time-stamped interaction networks without looking at the actual code [117] and can predict productivity and novelty generation [118,119]. Coordination costs increase with team size moderating productivity and innovation [120] a problem that managers need to solve. We will contact 10 companies including SME-s and large firms and work with their managers to understand how co-location can help collaborative interaction, and how collaboration between coworkers from different social class takes place. An innovative survey API will be used to collect responses in an automated way by pushing cellphone alerts (https://www.octopus-research.hu/). The consent of each individual software developer will be asked to connect their hashed version control ID with the survey. An option to trace communication activity and GPS location will be also offered for the employer and for the employees. This combined data will be used to understand the role of places and co-location and dynamics of focused teamwork in collaborative coding at very detailed granularity.

Methodology: The dynamics and productivity of collaboration in coding will be mapped. By using the survey API, we will map additional layers of social interaction, in the workplace or from home-office within the team and investigate their role in coding productivity.

Output: This data collection and design will enable us to explain how and what mixture of co-location and home office can facilitate process innovation in software development [121]. Our new approach of data collection will allow for groundbreaking contributions in explaining the mechanisms of network spillovers through collaboration. We will prepare two manuscripts for publication in high rank management and interdisciplinary journals.

Ethical questions: Data will be collected only with the fully-informed consent of respondents. We are going to hash the IDs of respondents and the IDs of their friends. Data will be stored and processed in full compliance with GDPR.

Table 1. *Risk mitigation for Project 1.2*

Risks	Mitigation
Limited company participation	• We employ a team of a Data Collection Officer and subcontracted Illustrator to prepare convincing information materials and win trust of managers. • We are already discussing the project with 3 companies.
Limited individual consent	• In case developers do not give consent to use their communication or geo-location data, we will ask survey questions to map locations and daily communication with colleagues. • In case individuals do not give consent to match their version-control ID with the survey, we will instead analyze openly available repos stored on GitHub.

Wok Package 2: Urban mobility, firms, and networks of inequalities

We will exploit globally unique administrative data to uncover how spatial interaction and urban mobility influence inequalities through social network layers and firm behavior. This work package will include tests on alternative mechanisms of spatial social interaction other than observable information flows. The data are available for Sweden and Hungary; thus, we focus on Stockholm and Budapest. Risk mitigation strategy is presented at the end of the work package.

Project 2.1. Social networks, urban mobility, and individual income. Despite the assortative mixing in the urban fabric [90], commuting to work can facilitate connections across social classes [64]. However, whether the connections that one establishes around the workplace can function as mitigator of unequal distribution of wealth, is not clear. Major problems for identifying this relationship are three-fold. First, individual wage is a result of long-term investments into human capital. Second, social networks that have been established long ago (family, school or in previous workplaces) might matter more than current information flows because these can also influence self-selection into jobs [122]. Third, there might be meeting locations different from work and home that can facilitate social tie creation [96].

Data: We will collaborate with international colleagues who will provide access to aggregate mobility data generated from ca. 200 thousand cellphones in Stockholm from Cuebiq, a GPS-tracker company for the period 2019-2024. These data will be used to generate aggregate mobility and visitation matrices between neighborhoods of homes, workplaces, and amenities, without following individual activity. The matrices will be used to measure temporal colocation of individuals from different neighborhoods. To qualify third locations of meetings, we will both apply datasets that can be purchased from Google and datasets that are freely available, like OpenStreetMap.

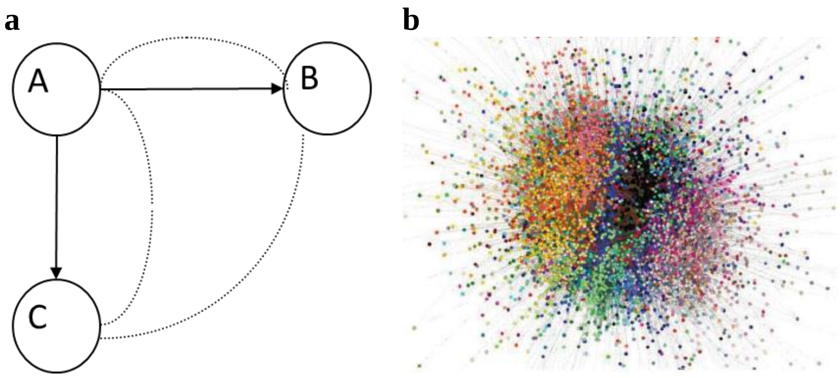

Figure 4. *Co-worker network creation from administrative data.* ***a)*** *Social networks (dotted lines) that span company borders (here A, B, and C) are generated by labor mobility (arrows)* [123]. ***b)*** *Co-worker networks representing entire Sweden aggregated to industries and regions (colors depict regions)* [32].

The mobility and location data will be matched on neighborhood level with the world class ASTRID dataset that contain the entire workforce in the city over the 2000-2022 period and is extended yearly. That is, at the end of 2026, we will have access to data until 2025. The dataset is created by the Statistical Office of Sweden and is made available for research with a year lag. Access will be provided to us at Umea University, Sweden that our colleagues will visit on a regular basis. Among others, the dataset contains very detailed individual information regarding careers, home location, and salary of everyone.

Methodology: We will construct co-worker-, family-, and classmate networks of every individual from the ASTRID data [32,123] (like in Figure 4). These networks will be used to explain differences in individual wage dynamics with panel-, and network regression techniques. Next, we construct commuting matrices from the home and work neighborhoods of individuals and correlate it with the mobility matrices. Then, we aggregate the social networks to the neighborhood level and analyze the dynamics of average neighborhood income in relation to the income in connected neighborhoods. Finally, we link neighborhoods with the mobility matrices to capture if citizens from two neighborhoods visit the same amenities. This will enable us to identify the type of co-location that can explain income convergence.

Output: The project will help us better understand the direction of the causal link between urban mobility, social networks, and income dynamics. We will be able to differentiate the role of various types of social connections from urban mobility in the dynamics of urban inequalities. Two manuscripts will be prepared for publication in high rank economic geography and interdisciplinary journals.

Ethical questions: Mobility data will be in an aggregate form to ensure that individuals are not identifiable. Data will be stored and processed in full compliance with GDPR.

Project 2.2. Urban mobility, firm mechanisms, and inequality. Individual income is created in firms that combine human capital with other inputs in production. In the urban mobility context, commuting mixes employees from various neighborhoods and potentially from various social classes providing the opportunity for knowledge sharing and income convergence. Yet, labor division within firms is decisive in income inequalities [124] while other operations of firms, such as trade with others, can also influence disparities through learning across firms [125]. New firm level data that contains employee characteristics, detailed records from balance sheets and transactions with other firms open the floor to disambiguate these mechanisms. In this project, we will aim to separate the role of commuting to work from the complex nature of firm production and trade in producing urban inequalities.

Data: Working with Hungarian Telekom that covers 60% of the Hungarian population, we will identify home, work, and third most visited locations of users in Budapest. Then, we acquire their data in an aggregate 100m×100m spatial matrix containing the number of present individuals by the identified location types and socio-demographic characteristics on an hourly basis for 18 months in 2020-2021. Next, we acquire an aggregate origin-destination matrix that connects neighborhoods of home, work, and third places in a privacy-saving manner. The social status of neighborhoods will be quantified by the average square-meter real estate prize from the database of the Hungarian Statistical Office.

The location and mobility matrices will be linked with precisely geo-located company data on aggregate 100m×100m level. These company data are processed and stored at the Host Institution of the project. The data include a linked employer-employee dataset that contains occupation, education background and wage of individuals. This data is available from 2013 and is updated on a yearly basis. Further, we will use a globally unique firm-level input-output network representing the universe of trade between Hungarian companies. This data is constructed from the VAT reports submitted to the Hungarian Tax office, it contains value of trade between companies, is available from 2016 and is updated yearly.

Methodology: We are going to analyze the dynamics of wage inequality within firms and explain it by the distribution of human capital, occupation structure, mixing of individuals from neighborhoods of different status, and trade networks. We investigate whether location of firms in the city is related to wage inequalities and aim to identify spillovers on inequality through trade linkages and shared locations of employee visits. Next, we analyze how mobility restrictions over COVID-19 lock-downs impact companies in terms of office and plant visits to identify the role of co-location in mitigating unequal wage distribution within and across companies.

Output: The analysis will provide new understanding on how mixing of social classes at workplaces shape wage inequalities and how this can spread to other companies as well. We will prepare two manuscripts for publication in high rank economics and interdisciplinary journals.

Ethical principles: Mobility data will be purchased from Hungarian Telekom in an aggregate form to ensure that individuals are not identifiable. Data will be stored and processed in full compliance with GDPR.

Table 2. *Risk mitigation for Work Package 2*

Risks	**Mitigation**
Too few mobility observations	•We will aggregate data on a higher spatial level and in bigger timeframe.
Matching firms and mobility fail	•We will apply probabilistic matching of companies to mobility data.
Data publishing issues	•Only aggregate data will be published.

Work Package 3: Neighborhoods, mobility, and outcomes of social interaction

Mobility and social media data will be used to explore how the type, content, and dynamics of interaction is related to income and innovation trends across neighborhoods in cities. We aim to investigate causality between mobility, interactions, and spatial outcomes. Due to data availability, our focus will be on Amsterdam, London, and the metropolitan area of large US cities.

Project 3.1. Mobility and social interaction in cities. In our previous research, we have shown that physical barriers, such as rivers, railroads, or major roads make social networks fragmented in cities that increases income inequalities [55]. Such barriers can create silos for social connections due to low level of access [86]. Despite mobility can facilitate mixing of social classes [64,126], whether improving access across neighborhoods can indeed amend broken social relations, is still unknown. We will analyze the causal relation between urban mobility and the type of interactions across neighborhoods and across business areas. This will be done by looking at the dynamics of mobility across neighborhoods before and after major infrastructure construction, and its impact on online communication.

Data: Based on previous experience [127], we have collected the most comprehensive geo-located Twitter database so-far in Amsterdam, London and New York City by focusing on those users who apply the automatic geo-tagging for their tweets. The Twitter Academic API enabled us to do this in a retrospective manner and download the public posts of the users. As in previous research, we will identify home and work neighborhoods of individuals, construct their mutual followership networks, and label the content they share and like (Figure 5).

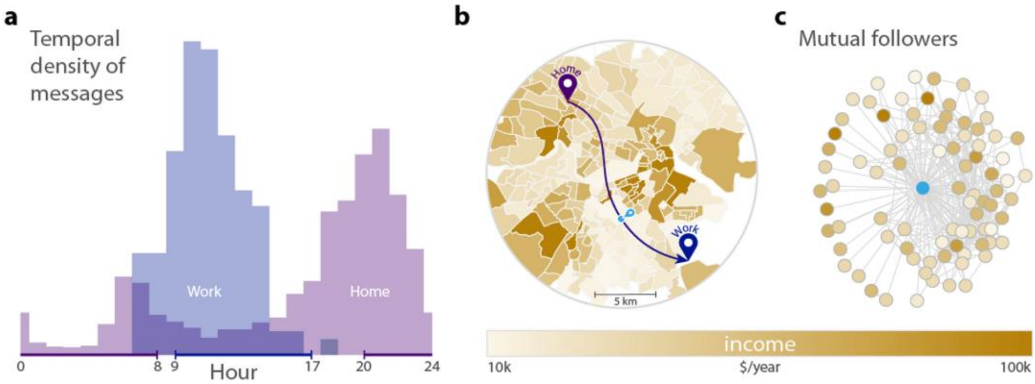

Figure 5. *Geo-location and spatial social networks from Twitter data* [64]. ***a)*** *Home and work by time of tweets.* ***b)*** *Commuting by the location of home and work.* ***c)*** *Social networks from mutual followership.*

We will collaborate with international colleagues who will provide access to aggregate and privacy-preserving urban mobility matrices from Cuebiq, a GPS-tracking company, that can capture the dynamics of mobility across neighborhoods (census tracts in the US, middle- or lower-layer output areas in London and ‘buurten’ in Amsterdam). We will investigate 6,607 bridge constructions and renovations in the 2016-2020 period in the US states that host the 15 largest metropolitan areas (data at LINK). Public transport stops and schedules can be used to estimate access in London (LINK) and in Amsterdam (GTFS LINK). Further open access data on amenities (from Google and OpenStreetMap) will be used to characterize neighborhoods' attraction. Finally, we will collect open access data that represent home income and demography in the cities analyzed (USA LINK, London LINK, Amsterdam LINK).

Methodology: We will identify how changes in urban access between locations influence mobility and hence change social interaction. We will focus on new follower and mention links on Twitter and will classify posts by using text mining or NLP (natural language processing) techniques to label the content of Twitter posts and investigate the changes in the type of information that flows between places.

Output: The analysis will provide previously unprecedented insights into the relationship between urban access, mobility, and social network formation. We will be able to identify the role of improving (or worsening) transport connections in facilitating online communication between places. Finally, we will explore the type of information that flows in social networks and across similar or different neighborhoods of cities. We will prepare two manuscripts for publication in high rank economics and interdisciplinary journals.

Ethical principles: All data used in this project will be stored and processed in full compliance with GDPR. Mobility data will be used in an aggregate form to ensure that individuals are not identifiable. We cannot ask consent from all Twitter users who post geo-located tweets. However, we will create a project website that we will circulate widely. Through this website, Twitter users could check if they are part of the sample and can request us to remove their data.

Project 3.2. Social networks, inequalities, and innovation in cities. Information flow in social networks is key for economic and technological progress. A dominant view suggests that cities thrive because they facilitate face-to-face interaction that is important for complex knowledge sharing [63] or can establish trust for distant collaborations [87]. Social networks in cities are known to be fragmented into spatial communities of nearby neighborhoods [60]. In this project, we aim to close major gaps by pioneering the investigation of how information flow in urban social networks impacts urban innovation and convergence/divergence of income across neighborhoods. In this project, we will analyze the role of mobility and online interaction in the income and innovation dynamics of neighborhoods of the investigated cities.

Data: We will continue to work with the datasets that we collect in Project 3.1 and use the mobility and online interaction networks across neighborhoods in London, Amsterdam, and New York City. Special attention will be paid to the mean and standard deviation of income in the city neighborhoods. This is available on a yearly basis in Amsterdam from 2004 until 2021, in London in a 2-year frequency from 2012 until 2018, and in the US cities on a yearly basis from 2009 to 2019.

Next, we use the publicly available data sources of the European and US Patent Offices to geo-locate the neighborhood of inventors. Patent data sources include the addresses of inventors that can be geo-coded with high precision [128,129]. This will enable us to link patent information to the neighborhoods in the cities we analyze, based on the inventors' addresses. These rich data sources include co-inventors, technology classes, citations, and description of the patent.

Methodology: We will predict income dynamics with variables extracted from the neighborhood-level interaction networks and compare predictions across cities. Special attention will be given to the spatial patterns and types of information content explored in Project 3.1. We will use different layers of these interaction networks to describe divergence and convergence of income across neighborhoods. Next, we construct spatial collaboration networks from patents and correlate these with the Twitter interaction network. We will investigate patenting dynamics, including the number of new patents and the level of their novelty, in neighborhoods based on the collaboration network and on the social interaction network.

Output: Our analysis will uncover the relation of neighborhood-level dynamics of inequality and patenting with urban mobility and social interaction networks. This pioneering work will include many cutting-edge contributions to the literature including the causal link between the content of spatial social interaction in urban income divergence/convergence, and the social interaction approach to within city dynamics of innovation. We will prepare two papers for publication in interdisciplinary, economics, or innovation studies journals.

Ethical principles: We will follow identical steps as described in Project 3.1

Table 3. *Risk mitigation for Work Package 3*

Risks	**Mitigation**
Misclassification of content	• Many classification strategies will be applied to detect content of interactions. Results of these will be cross-checked to increase robustness of findings.
Time period of data limits causal inference	• We will use home and work location from Tweets to estimate commuting routes instead of actual mobility. • We will instrument spatial social interaction with changes in urban access.
Too few patents in neighborhoods of inventors' home	• We will aggregate neighborhoods to larger spatial units in the city. • We will use assignee locations and generate spatial interaction networks across workplaces instead of home neighborhoods. • Aggregate dynamics on the city level. Extend the number of US cities to 50 as in [64].

Work Package 4: Diffusion, scientific advance, and inequalities

We will access large open-access publication databases and analyze how dynamics of co-author networks, and diffusion of novel information on them are related to the creation of novelty and unequal trends of prestige in and across cities.

Project 4.1. Diffusion of information and new knowledge creation. Diffusion of information and innovation in social networks follow the mechanisms of social influence: individuals are more likely to accept an idea or adopt a novelty when it is already well adopted among their network neighbors [56,103,130]. Science is no exception: most incremental progresses circulate in specialized communities [131] that diverge in knowledge and methods, hence, are difficult to bridge [132]. Yet, breakthrough ideas can diffuse across fields yielding cross-fertilization of disciplines and general progress [133]. In this project, we aim to disentangle the social influence mechanism prevailing in co-author networks from other types of knowledge diffusion and analyze the role of such diffusion processes in new knowledge generation.

Data: Two large and open-access publication databases will be used in this project. First, the PubMed Knowledge Graph covers from 532K authors and their papers in medicine and life sciences. The PubMed Knowledge Graph contains employment history from ORCID [134] for the 1913-2018 period, including precise geo-location of affiliations. It also contains rich information about the scientific articles, including keywords, and further entities (eg. genes, diseases, drugs, species, mutations) identified in their abstracts and their position in the abstract. Second, Scopus is the world's largest citation database and provides access to their data through an API. This dataset covers many disciplines, author information (eg. affiliation) and publication details (journal, abstract, citations). The data goes back to the 18th century.

Methods: We will construct dynamic co-authors networks within and across cities, in which the spatial dimension will be given by affiliations. New occurrences of keywords, and expressions in abstracts will be identified as novel information. We will track how new concepts spread in geographical space and in the network and how spreading patterns change over decades. Special attention will be paid to characterizing the circulation of new information within and across communities of co-author networks. We will analyze divergence and convergence of knowledge domains by classifying abstract contents and measuring their similarity. Finally, we will analyze how bridging across communities yield new combinations of knowledge domains and how this impacts research.

Output: The results will provide us with new understanding on how bridging in spatial social networks can yield diffusion of scientific ideas and how boundary-spanning flows facilitate the creation of new knowledge. We will prepare two papers for publication in interdisciplinary, and innovation studies journals.

Ethical principles: The PubMed database poses no ethical issues. The Scopus API is restricted and does not allow the download of the entire database. Accordingly, we will follow the rules of the data download and concentrate on selected disciplines.

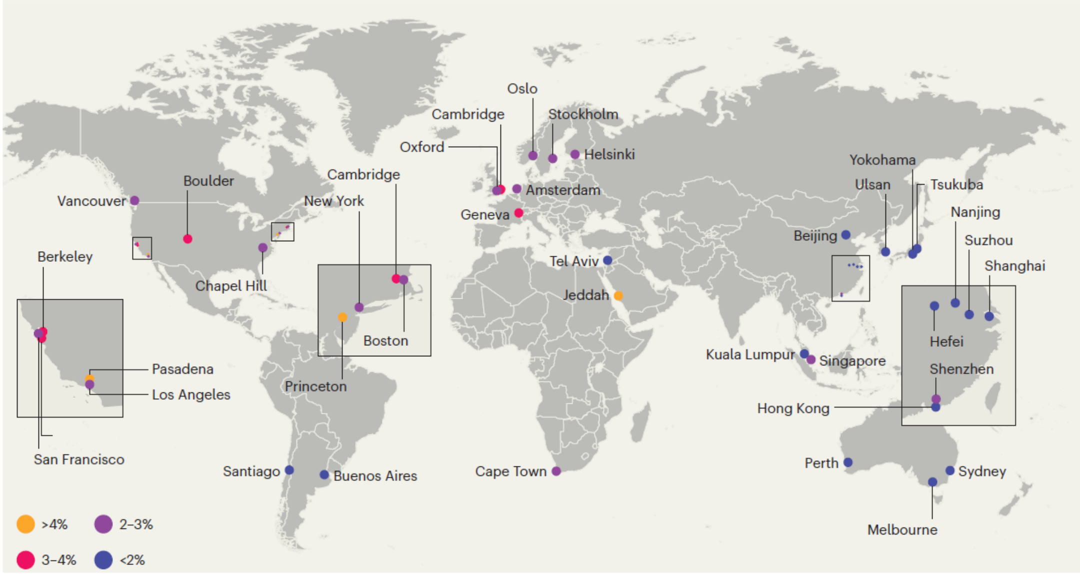

Figure 6. *Prestige of science cities* [135]. *The ratio of highly cited papers shows continental differences.*

Project 4.2. Individual career, life-cycle of ideas, and inequality in impact. Scientific prestige and impact concentrates in geographical space (Figure 6). Location can be decisive for impact differences since global centers attract and keep talent [136,137] and put scientist to the core of collaboration networks [138,139]. Scientific ideas have a life-cycle and the core science cities and networks dominate the selection and validation of ideas [140]. Spillovers from outstanding scientists to collaborators or faculty can boost recognition through networks [141]. Yet, the recognition of ideas largely depends on the combination of conventional knowledge and novelty [113], stressing the role of competition of ideas and their life-cycle [142]. This project will address how individual recognition depends on location, spatial mobility, co-author networks and the adoption of ideas over their life-cycle [142].

Data: We will continue to work with the data collected in Project 4.1.

Methods: We will look at individual scientists and quantify the degree of inequality in impact, measured by citations, within and across cities in a cross-disciplinary comparison. Then, we measure how this inequality can be reduced by adopting new concepts, establishing new co-author relations, and changing affiliations. We will focus on co-author networks within cities and quantify the local effect of outstanding, mobile, or gatekeeper scientists on their local co-authors. We will use the external variation of temporality in political divide across countries (eg. the Eastern and Western block during Cold War) and the opportunities of research visits and collaborations across borders and distances after such divides disappear.

Output: Results will provide new insights into how mobility, new collaborations and entering new or old fields can help individuals to achieve impact and how central or peripheral locations can benefit from these mechanisms. We will prepare two papers for publication in interdisciplinary, and innovation studies journals.

Ethical principles: Same applies as in Project 4.1

Table 4. *Risk mitigation for Work Package 4*

Risks	**Mitigation**
Infrastructure demand	• A Data Engineer will be employed to set up an effective storage and querying workflow.
Too many novel information, misclassification of novelty	• Seek advice from experts to characterize the keywords and abstract elements we select for modeling.
Scopus API is restricted	• In case Scopus API does not allow for a large data download, we will focus on the PubMed data only

Work Package 5: Innovation and inequalities in evolving networks

The Principal Investigator will summarize the empirical findings into a unified theoretical framework.

Project 5.1. Theoretical framework development. In the last year of the project, a common theoretical framework will be developed. This framework will leverage the empirical findings developed in the empirical work packages.

Methods: Summarizing and comparing empirical findings into a common frame that explains how different mechanisms of social network evolution can lead to innovative outcomes and to diverging/converging income, prestige and knowledge in social and collaboration networks.

Output: A book will be written with the title “Innovation and inequalities in evolving spatial networks”.

III.2 Yearly Breakdown of Tasks in the First 3 Years

Year 1

General tasks – Team establishment

- Task G1: Setting up the Advisory Board – We will invite internationally recognized scholars to join the Advisory Board of the project.
- Task G2: Establish the brand of the project – A new website and logo will be created to establish a brand that is easy to communicate in academia, with industrial and government partners and with the general public.

Work Package 1 – Preparation for data collection

- Task 1.1: Consultation with companies – The PI, the Senior Researcher and the Postdoctoral Researcher working on WP1 will contact company managers with a research proposition and adjust specific research questions to the problems these managers face.

- Task 1.2: Survey questionnaire development – We will consult with the Octopus API team (Centre for Social Sciences) and develop a survey questionnaire in coordination with the managers of the analyzed companies.
- Task 1.3: Github data platform development – We will use existing software packages to develop our algorithmic approach of generating dynamic collaboration networks of software engineers in companies.

Work Package 2 – Data collection and processing
- Task 2.1: Start of the mobility data purchase – We will purchase aggregate mobility data from Hungarian Telekom.
- Task 2.2: Mobility data processing – The mobility data available through our collaborators will be connected to the Swedish ASTRID database and to the Hungarian ADMIN database.
- Task 2.3: Additional data collection – We collect additional information from open-source databases on the neighborhood-level income and amenity portfolio.

Work Package 3 – Data collection and processing
- Task 3.1: Twitter data download and characterization – The Data Engineer Team will process the downloaded geo-located tweets from Amsterdam, London, and New York City for the 2010-2022 period and generate dynamic spatial interaction networks in these cities.
- Task 3.2: Mobility data collaboration – We scale up the existing collaboration with international partners who have access to mobility data in the analysed cities.
- Task 3.3: Administrative income data and patent data – We collect additional information from open-source databases on the neighbourhood-level income and amenity portfolio.

Work Package 4 will be started in Year 3 to balance budget constraints. Work Package 5 starts in Year 5.

Year 2
Work Package 1 – Data collection, and descriptive reports
- Task 1.4: Survey and software code data collection – We collect the data with the survey application and from the GitHub repositories of companies. These data will be merged keeping the anonymity of respondents that will provide unique opportunities to analyse co-location, dynamic collaborations, and individual and collective outcomes.
- Task 1.5: Descriptive analysis – The collected data will be analysed in a descriptive manner.
- Task 1.6: Reports to company managers – We write short technical reports to provide insights for the respective companies.

Work Package 2 – Urban mobility and social networks from administrative data
- Task 2.4: Finish the mobility data purchase
- Task 2.5: Social network analysis – A descriptive and modelling analysis will be run to investigate the relation of urban mobility with different layers of social networks generated from administrative data sources.
- Task 2.6: Paper preparation 2.1 – We prepare a manuscript on the descriptive analysis of the relationship between urban mobility and social networks at the neighbourhood level and submit to Urban Studies or similar outlets.

- Task 2.7: Paper preparation 2.2 - We prepare a manuscript on the relationship between urban mobility, social networks, and the individual-level inequality and income dynamics and submit to Nature Communications or similar outlets.

Work Package 3 – Urban mobility and social interaction

- Task 3.4: Urban accessibility analysis – We analyse how the changes in urban accessibility influences the extent and features of social interaction and collaboration across neighbourhoods in the analysed cities.
- Task 3.5: Paper preparation 3.1 – A manuscript on the relationship between urban accessibility, mobility, and the content of social interaction across neighbourhoods and will be prepared and submitted to EPJ Data Science or similar journals.
- Task 3.6: Paper preparation 3.2 – A manuscript on the relationship between urban accessibility, mobility, and scientific and innovative collaboration will be prepared and submitted to Research Policy or similar journals.

Work Package 4 will be started in Year 3 to balance budget constraints. Work Package 5 starts in Year 5.

Year 3

Work Package 1 – Modelling and scientific paper

- Task 1.9: Modelling – Statistical, and simulation models will be applied to analyse the data with scientific rigorousness.
- Task 1.10: Paper preparation 1.1 – A manuscript will be prepared for publication on the role of co-workers' co-location in the productivity of collaborations. We will submit it to Social Networks or similar outlets.
- Task 1.11: Paper preparation 1.2 – A manuscript will be prepared on the dimensions of inequalities in dynamic collaboration networks in companies and will be submitted to Nature Human Behaviour or similar outlets.

Work Package 2 – Urban mobility, firm behaviour, and firm networks

- Task 2.7: Firm production and networks – We investigate the relationship between the patterns of commuting and visitation of urban amenities with firms' productivity, the complexity of production, and the trade networks between firms.
- Task 2.8: Paper preparation 2.3 – A manuscript on commuting, meetings in the city and firm production will be prepared and submitted to Journal of Urban Economics or similar outlets.
- Task 2.9: Paper preparation 2.4 – An interdisciplinary manuscript on the nature of wage inequalities in firms, its spillovers across firms and the transmitting effect of urban mobility, and firm trade will be prepared and submitted to Science Advances or similar outlets.

Work Package 3 – Publication of results

- Task 3.7: Paper preparation 3.3 – A manuscript on urban mobility, social interaction and income divergence will be submitted to PNAS or similar journals.
- Task 3.8: Paper preparation 3.4 – A manuscript on urban mobility, collaboration networks and scientific-, innovative output in cities will be submitted to Economic Geography.

Work Package 4 – Data collection and descriptive analysis

- Task 4.1: We hire a new full-time international postdoc for 2.5 years to deal with diffusion of scientific novelty and inequalities in scientific impact.
- Task 4.2 Data download and processing – The scientific publication and patent databases will be downloaded and cleaned for spatial analyses.
- Task 4.3: Descriptive analysis – An explorative descriptive study will uncover how the spatial inequalities in scientific impact changed over time and across disciplines.

Work Package 5 starts in Year 5.

Expected outcome in the first 3 years and dissemination of results

We will prepare at least 10 open-access papers (detailed above) for publication in leading interdisciplinary journals and in top field journals of economic geography and regional science, urban economics, urban studies, social networks, management, and innovation studies. As Work Package 4 will start in Year 3, the 4 publications planned in that WP regarding spatial inequality and spillovers of scientific impact and the spatial diffusion of novelty will be prepared in Years 4 and 5. Finally, a book entitled "Innovation and inequalities in evolving spatial networks" is planned to be prepared in Year 5.

The first three years of the project will devote significant efforts to data collections. We will publish the collected data in a form that does not violate GDPR and national law. We will publish code on GitHub.

Four online workshops will be held to disseminate results and to start new collaborations and collect further ideas. Every team member will be given the opportunity to attend one international conference per year.

We will disseminate knowledge to the scientific community on social media channels and inform society frequently through public media appearances.

III.3 Resources

The current Lendület research proposal aims to continue what the PI and the research team has started within the Agglomeration and Social Networks Lendület Research Group (ANET Lab). The ANET Lab has gained 150 thousand Ft support in 2017. We published 35 high impact articles over the reporting period (cumulated IF = 200.288) and were very active in teaching and dissemination of the results to the wider public. ANET Lab has received "excellent" evaluation from the President of the Hungarian Academy of Sciences in 2022.

Furthermore, we have been invited to establish a new research group at Corvinus University of Budapest. The Laboratory for Networks, Technology and Innovation (NETI Lab) has been an outstanding unit at the university and contributed immensely to the grant winning "Collective Learning Data Lab" ERA-Chair application to establish a new research group of the Center for Collective Learning (CCL) with the leadership of Professor César Hidalgo.

Today, our community including NETI Lab and CCL is growing and attracts international scholars as well as postdocs to join us in Budapest and we help talented Hungarian students to start their research activities. However, many from the community who contributed to the development of ANET Lab are working now abroad and the aim of this current Lendület H proposal is to bring these very talented researchers back to Budapest and integrate them again into our growing community. In this section, I explain the capacities of the PI, the Research Team, the Host Institute, and main international collaborators. The specific aim is to motivate that the continuation of the ANET Lab Lendület funding will facilitate the developments of a growing, internationally recognized, but still young research community in network science, economic geography, urban studies, computational social science, and related fields.

Principal Investigator

Balázs Lengyel is a Senior Research Fellow at the Centre for Economic- and Regional Studies and an Associate Professor at the Corvinus University of Budapest. He is an internationally recognized economic geographer pioneering the intersection between economic geography and complex social networks. Using administrative data, he introduced large-scale co-worker network analysis into economic geography [32,123]. His project on the geography of online social networks [86] was recently exhibited in the "Álmok Álmodói" exhibition among top achievements of Hungarian science (exhibition, video) and led to breakthrough contributions on spatial diffusion of innovation [110] and networks of urban inequalities [55]. He is a lecturer at the Oxford Summer School on Economic Networks (2020 and 2023) and was invited to publish in a Nature supplement in 2020 [135]. Research visits at MIT (2016-2017), Umea University (2014-2015), Utrecht University (2012), and Aalborg University (2008) illustrate his wide international collaborations. He has 61 journal articles, out of which 46 were published in international journals. According to MTMT, he received 1233 independent citations to his works (H index is 21), on Google Scholar, he has 2130 citations (H index is 27).

Balázs has strong experience in leading research teams and scientific communities. Being the head of ANET Lab from 2017 and co-director of NETI Lab from 2020, he is directing four postdocs, four PhD students, and three research assistants. The alumni of his team have positions at top universities including University of Amsterdam, Northeastern University London, and Complexity Science Hub, Vienna. He has been a vice-president of the Regional Science Committee of the Hungarian Academy of Sciences and board member of the Hungarian Association of Regional Science. Balázs is a founding member of the Hungarian Young Academy (HYA) and has coordinated the main research project of HYA that mapped the work conditions of early career researchers in Hungary (LINK TO REPORT). Balázs has experience in ethical matching of identifiable social media and publication database with sensitive survey data [143]. He was a recipient of the Lendület (2017), Bolyai (2017), Eötvös (2016), and Rosztoczy (2015) awards. He has been the PI of five Hungarian research grants (total sum around 700K EUR) and has been a main contributor to the "Collective Learning Data Lab" ERA-Chair grant application that was awarded 2.5 Million Euro to the Corvinus University of Budapest.

Team, tasks, and skills

Most of the skills required for the project have been accumulated in ANET Lab and NETI Lab over the years. We collected lot of experience in large data collection and processing (including social media, administrative data, mobility data, surveys etc.). We have managed sensitive data collection campaigns, coordinated legal preparations, and data infrastructure development. The two groups form an interdisciplinary community that has been a home for economic geographers, economists, geographers, sociologists, data scientists, physicists, and mathematicians. The growing group of CCL

The aimed research team of the current project will consist of nine further members: five Senior Researchers, and four Postdocs. In addition, we will recruit two PhD students at Corvinus University (no tasks are assigned in the Work Plan and no budget allocated) and hire the Data Engineer team of KRTK Data Bank for their part time contribution.

Senior researchers: Johannes Wachs is an associate professor in data science at Covrinus University Budapest and an internationally recognized expert in the field of open source software collaborations. Zoltán Elekes is a researcher in the field of economic geography at the Centre for Economic- and Regional Studies and a grant holder from the Swedish National Research Fund at Umea University. Orsolya Vásárhelyi is a sociologist and an assistant professor of social data science at Corvinus University where she works as an expert of science of science. Rebeka O. Szabó is an assistant professor at Corvinus University and an emerging international expert in organizational behavior. Sándor Juhász is a Marie-Curie Fellow at the Complexity Science Hub Vienna and is an emerging international expert of urban mobility and supply chain networks.

The Senior Researchers will be employed part-time for five years fixed-term, for the reasons of saving money for the development of the Postdocs. They will serve as consultants of Work Packages and will engage actively with Work Packages 1, 2, 3, 4, and 5.

Postdoctoral researchers from ANET: Gergő Tóth is an economist, is a postdoctoral researcher at Umea University and has published intensively in the field of spatial social networks. He has received the Junior Prima Prize in 2022. Gergő Pintér is a postdoctoral researcher at Corvinus University where he works on urban mobility topics. Eszter Bokányi is a postdoctoral researcher at University of Amsterdam where she works on topics of population-scale social networks. These colleagues will work on Work Packages 1, 2, and 3. The Postdocs will be employed full time over the entire project. In Year 3, we will hire Luca Gallo for a 2.5 years full time job or part-time research fellow, to concentrate on Work Package 4. Luca is a postdoctoral researcher at the Central European University. The four Postdocs concentrate on one work package each, will be responsible to design the research together with the PI, conduct analysis, and write the paper with the PI.

The PI will direct the group that will include all managerial tasks. He will lead all work packages – with the help of the Senior Researchers – and will devote extra focus on Work Package 5.

Host Institution

The Institute of Economics at the Centre for Economic and Regional Studies is the number one academic organization in Hungary, according to the REPEC database. The Institute has strong traditions in labor economics, health economics, and education economics.

Collaborations and Advisory Board

Our team will ask advice and data access from collaborators. Prof. Rikard Eriksson from Umea University will provide access to the ASTRID Database on Swedish employees. This data can be only worked with in Umea. Prof Riccardo Di Clemente from Northeastern University London will collaborate with us on mobility data and by running our code developed from our own Budapest data, will scale up our analysis by including major European and US cities. Bence Ságvári from ELKH Budapest is the leader of the survey application development that the project will use and has agreed to consult our survey campaign. In the first six months of the project, we will set up a board of internationally recognized scholars (5-8 members) in economic geography, urban data science, urban economics, urban studies, social network analysis, and innovation studies to serve as inspectors of scientific excellence.

Budapest, 2024.02.06. Balázs Lengyel

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143. Lőrincz L, Chihaya GK, Hannák A. Global connections and the structure of skills in local co-worker networks. Appl Net Sci. 2020;5: 78.

IV. Personal background of the planned research project

CURRICULUM VITAE

Principal Investigator

Surname: Lengyel
First name: Balázs
Date of birth: 10.15.1980.
ORCID: 0000-0001-5196-5599
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Current positions:

2023-: Associate Professor at the Data Analytics and Information Systems Institute, Corvinus University of Budapest

2020-: Senior Research Fellow at Laboratory of Networks, Technology and Innovation, Centre for Advanced Studies, Corvinus University of Budapest

2018-: Senior Research Fellow at the Economics of Networks Research Unit, Centre for Economic and Regional Studies, Hungarian Academy of Sciences, Budapest

2017-: Head, Agglomeration and Social Networks Lendület Research Group, Hungarian Academy of Sciences, Budapest

Previous positions:

2023: Lecturer at Rajk Special College on "Network Science".

2020: Lecturer at Széchenyi István Special College on "Network Science".

2014-: External lecturer at the Department of Economics, ELTE University on "Regional Economics" and "Network Science".

2012-: External lecturer at the Faculty of Economics and Business Administration, University of Szeged on "Knowledge Management".

2013-2017: Research fellow at the Economics of Networks Research Unit, Centre for Economic and Regional Studies, Hungarian Academy of Sciences, Budapest

2010-2018: Research fellow at the International Business School Budapest

2008-2010: Research associate at the Institute for World Economics, Hungarian Academy of Sciences

2006-2007: Research assistant at the Centre for Regional Studies, Hungarian Academy of Sciences

2004-2005: Strategy assistant at the National Office for Research and Technology, Hungary

URL for web site: https://anet.krtk.mta.hu/researchers/b-lengyel/

Education and key qualifications:

2010: PhD in Economics and Organizational Studies on the thesis "Spatial analyses of the Hungarian knowledge-based economy – regional innovation systems and knowledge-base". Doctoral School of Economics and Management Sciences, Budapest University of Technology and Economics, Hungary. Supervisor: Gábor Papanek.

2004: MA in Economics and Business Administration. Faculty of Economics, University of Szeged, Hungary

Research experience:

August 2016 – July 2017: 12 months stay as a visiting scholar at Human Mobility and Networks Lab, Massachusetts Institute of Technology, USA

September 2014 - June 2015: 4 months stay as visiting research fellow at the Department of Geography and Economic History, Umeå University, Sweden

September – November 2008: 3 months visiting PhD student at Department of Business Studies, Aalborg University

Received fellowships and grants:

2022: Corvinus Research Excellence Award

2021: 45M HUF (121K EUR), from the National Office for Research and Technological Innovation for the project "Mobility, social- and economic networks, and inequalities in cities over the pandemic" (2021-2025)

2018: 20M HUF (54K EUR), from the National Office for Research and Technological Innovation for the project "Spatial Diffusion of Innovation" (2018-2020)

2018: 26.4M HUF (71K EUR), from Hungarian Scientific Research Fund for the project "Social and Collaboration Networks and Individual and Organizational Performance" (2018-2021)

2017: 150M HUF (404K EUR), MTA Lendület, "Agglomeration and Social Networks". (2017-2022)

2017: Bolyai Scholarship of the Hungarian Academy of Sciences (resigned)

2016: Eötvös Scholarship of the Hungarian State

2015: Rosztoczy Foundation Award

2012: 1.7M HUF (5K EUR) from Hungarian Scientific Research Fund for the project "Regional economic growth in a dual economy: evolutionary and complexity perspectives". (2012-2014)

2012: Excellent Young Researcher Award from the Hungarian Regional Science Association

2003: Research Award from the City of Szeged

2002: Management Education Fund Award, Hungary

The bibliographic data and the number of independent citations of the 5 most important publications over the last five years:

Kogler DF, Whittle A, Kim K, **Lengyel B** (2023) Understanding Regional Branching: Knowledge Diversification via Inventor and Firm Collaboration Networks. *Economic Geography* 99 (5): 471-498. DOI: 10.1080/00130095.2023.2242551. Independent citations: 1

Abbasiharofteh M, Kogler DF, **Lengyel B** (2023) Atypical combination of technologies in regional co-inventor networks. *Research Policy* 52. DOI: 10.1016/j.respol.2023.104886. Independent citations: -.

Tóth G, Wachs J, Di Clemente R, Jakobi Á, Ságvári B, Kertész J, **Lengyel B** (2021) Inequality is rising where social network segregation interacts with urban topology. *Nature Communications* 12(1): 1143. DOI: 10.1038/s41467-021-21465-0. Independent citations idézők: 75.

Csomós Gy, Vida Zs, **Lengyel B** (2020) Science cities seek new connections. *Nature* 585(7826): S58-S59. DOI: 10.1038/d41586-020-02579-9. Independent citations idézők: 1.

Elekes Z, Boschma R, **Lengyel B** (2019) Foreign-owned firms as agents of structural change. *Regional Studies* 53 (11): 1603-1613. DOI: 10.1080/00343404.2019.1596254. Independent citations: 70.

The bibliographic data and the number of independent citations of further 5 significant publications:

Eriksson R, **Lengyel B** (2019) Co-worker networks and agglomeration externalities. *Economic Geography* 95(1): 65-89. DOI: 10.1080/00130095.2018.1498741. Independent citations: 19.

Juhász S, **Lengyel B** (2018) Creation and persistence of ties in cluster knowledge networks. *Journal of Economic Geography* 18(6): 1203-1226. DOI: 10.1093/jeg/lbx039. Independent citations: 33.

Lengyel B, Eriksson R (2017) Co-worker networks, labour mobility and productivity growth in regions. *Journal of Economic Geography* 17(3): 635-660. DOI:10.1093/jeg/lbw027. Independent citations: 15.

Lengyel B, Varga A, Ságvári B, Jakobi Á, Kertész J (2015) Geographies of an online social network. *PLoS ONE* 10(9): e0137248. DOI: 10.1371/journal.pone.0137248. Independent citations: 62.

Lengyel B, Leydesdorff L (2011) Regional Innovation Systems in Hungary: the failing synergy at the national level. *Regional Studies* 45(5): 677-693. DOI: 10.1080/00343401003614274. Independent citations: 90.

CURRICULUM VITAE

Senior Researcher

Surname: Wachs
First name: Johannes
Date of birth: 10.21.1987.
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Current positions:

2023-: Associate Professor at the Data Analytics and Information Systems Institute, Corvinus University of Budapest

2023-: Senior Research Fellow at the Economics of Networks Research Unit, Centre for Economic and Regional Studies, Hungarian Academy of Sciences, Budapest

Previous positions:

2020-2023: Researcher at Complexity Science Hub Vienna, Austria

2020-2023: Assistant Professor at Vienna University of Economics and Business (WU)

2019-2020: Postdoctoral Fellow at RWTH Aachen University

URL for web site: https://johanneswachs.com

Education and key qualifications:

2019: PhD, Network Science, Central European University, Advisor: János Kertész.

2012: MSc in Applied Mathematics, Central European University.

2009: BS in Mathematics and Economics, Tulane University, USA.

Research experience:

February-April 2018: 3 months stay as a visiting PhD student at the Oxford Internet Institute, Oxford University. Host: Taha Yasseri

Received fellowships and grants:

2023: 30M HUF (78K EUR), from the National Office for Research and Technological Innovation for the project “Building blocks of the digital economy” (2024-2027)

2021: 16K EUR from the City of Vienna Jubilee Fund for the project “Geography of Open Source Software” (2021-2022)

2021: 18.5K EUR from the WU Projects Fund for the project “Open (Science) Data Usage” (2021-2022).

2020: Team member of 1st Place submission Public procurement corruption risks: Harnessing Big Data for better fiscal governance and growth at the IMF Anti-Corruption Challenge, 2020. $50K

2018: Honorable Mention for the 2018 Lijphart/Przeworski/Verba Dataset Award from the Comparative Politics Section of the American Political Science Association for *opentenders.eu.*

2016: Academic Achievement Award for First-Year Doctoral Students, CEU, 2016.

The bibliographic data and the number of independent citations of the 5 most important publications over the last five years:

Tóth G, **Wachs J**, Di Clemente R, Jakobi Á, Ságvári B, Kertész J, Lengyel B (2021) Inequality is rising where social network segregation interacts with urban topology. *Nature Communications* 12 1143. DOI: 10.1038/s41467-021-21465-0. Independent citations: 75.

Wachs, J., Nitecki, M., Schueller, W., & Polleres, A. (2022). The geography of open source software: Evidence from GitHub. *Technological Forecasting and Social Change*, *176*, 121478. Independent citations: 6

Wachs, J., & Vedres, B. (2021). Does crowdfunding really foster innovation? Evidence from the board game industry. *Technological Forecasting and Social Change*, *168*. Independent citations: 14

May, A., **Wachs, J**., & Hannák, A. (2019). Gender differences in participation and reward on Stack Overflow. *Empirical Software Engineering*, *24*, 1997-2019. Independent citations: 44

Wachs, J., & Kertész, J. (2019). A network approach to cartel detection in public auction markets. *Scientific Reports*, *9*(1), 10818. Independent citations: 27

The bibliographic data and the number of independent citations of further 5 significant publications:

Moldon, L., Strohmaier, M., & Wachs, J. (2021, May). How gamification affects software developers: Cautionary evidence from a natural experiment on GitHub. In *2021 IEEE/ACM 43rd International Conference on Software Engineering (ICSE)* (pp. 549-561). IEEE. Independent citations: 11

Wachs, J., Yasseri, T., Lengyel, B., & Kertész, J. (2019). Social capital predicts corruption risk in towns. *Royal Society Open Science*, *6*(4), 182103. Independent citations: 20

Gessler, T., Tóth, G., & Wachs, J. (2022). No country for asylum seekers? How short-term exposure to refugees influences attitudes and voting behavior in Hungary. *Political Behavior*, *44*(4), 1813-1841. Independent citations: 31

Wachs, J., Fazekas, M., & Kertész, J. (2021). Corruption risk in contracting markets: a network science perspective. *International Journal of Data Science and Analytics*, *12*, 45-60. Independent citations: 28

Schueller, W., Wachs, J., Servedio, V. D., Thurner, S., & Loreto, V. (2022). Evolving collaboration, dependencies, and use in the Rust Open Source Software ecosystem. *Scientific Data*, *9*(1), 703. Independent citations: 0

CURRICULUM VITAE

Senior Researcher

Surname: Orsolya
First name: Vásárhelyi
Date of birth: 07.26.1988.
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MTMT: 0000-0001-6326-06

Current positions:

2023-: Assistant Professor at the Data Analytics and Information Systems Institute, Corvinus University of Budapest

2023-: Assistant Professor at Center for Collective Learning, Centre for Advanced Studies, Corvinus University of Budapest

2023-: Postdoctoral Research Fellow at Laboratory of Networks, Technology and Innovation, Centre for Advanced Studies, Corvinus University of Budapest

2019-: External Research Fellow, LINK Lab, Northwestern University, IL, USA

2018-: Ambassador of the Women in Data Science Worldwide Organization

Previous positions:

2023 January - 2023 September: Postdoctoral Research Fellow, Central European University Democracy Institute, Hungary

2022 Maternity Leave

2021-2022: Postdoctoral Research Fellow at Laboratory of Networks, Technology and Innovation, Centre for Advanced Studies, Corvinus University of Budapest, Hungary

2020-2021: Postdoctoral Research Fellow, CIM, University of Warwick, UK

2019-2020: Visiting Researcher, Northwestern University, LINK Lab, IL, USA

2018 May-2018 August: Data Science for Social Good Fellow, University of Chicago & Universidade NOVA de Lisboa, Lisbon, Portugal

2017 May – 2018 January : Graduate Research Intern, Typeform, Barcelona, Spain

URL for web site: https://www.orsivasarhelyi.com/

Education and key qualifications:

2020: PhD in Network Science, Central European University, Budapest, Hungary

Thesis: Computational and relational understanding of gender inequalities in science and technology, Supervisor: Balázs Vedres

2012: MSc, Survey Statistics, Eötvös Loránd University

2010: BA in Social Sciences, Eötvös Loránd University

Received fellowships and grants:

2023: 10, 000 EUR Women and Science Chair, a Paris Dauphine-PSL University and its Foundation Chair, in partnership with Amundi, Fondation L'Oréal, La Poste, Generali France, Safran and Talan.

2022: Corvinus Research Excellence Award

2021: Corvinus Research Excellence Award

2019: WomEncourage ACM Grant, Rome, Italy

2019: Facebook WiDS MicroFellowship, Stanford,CA, USA

2019: Doctoral Research Support Grant Northwestern University, Chicago, IL,USA

2018: Data Science for Social Good Fellowship,Lisbon, Portugal

2017: WomEncourage ACM Grant, Barcelona, Spain

2017: Grace Hopper Scholar, GHC, Orlando, FL, US

2016: PyCon, Diversity Grant, Portland, OR, USA

2016: NetSciX Young Researcher Conference Grant, Tel-Aviv, Israel

2015: Bridge Budapest Fellowship, UAE, SouthKorea, India, Japan

2010: Erasmus Scholarship. Brussels, Belgium,2010

The bibliographic data and the number of independent citations of the 5 most important publications over the last five years:

- Vedres, B. & **Vásárhelyi, O.** (2023) Inclusion unlocks the creative potential of gender diversity in teams. *Scientific Reports* 13.1:13757. Independent citations: 4
- **Vásárhelyi, O**., Zakhlebin, I., Milojević, S., & Horvát, E. Á. (2021) Gender inequities in the online dissemination of scholars' work. *Proceedings of the National Academy of Sciences*, 118(39). Independent citations: 17
- Koltai, J., **Vásárhelyi, O**., Röst, G., & Karsai, M. (2022) Reconstructing social mixing patterns via weighted contact matrices from online and representative surveys. *Scientific Reports*, 12(1), 1-12. Independent citations: 6
- Natera, L., Deritei, D., Vancsó A., **Vásárhelyi, O.** (2020) Quantifying life quality as walkability on urban networks: the case of Budapest in *Complex Networks and Their Applications VIII*, Springer International Publishing. Independent citations: 8
- Vedres, B. & **Vásárhelyi, O.** (2019) Gendered behavior as a disadvantage in open source software development, *EPJ Data Science* 8 (1), 25. Independent citations: 13

The bibliographic data and the number of independent citations of further 5 significant publications:

- **Vásárhelyi, O.** & Horvát, E. Á. (2023) Who benefits from altmetrics? The effect of team gender composition on the link between online visibility and citation impact . Proceedings 19th International Society of Scientometrics and Informetrics Conference, Bloomington, Indiana Independent citations: 0
- Karsai, M., Koltai, J., **Vásárhelyi, O.** and Röst, G. (2020) Hungary in mask/maszk in Hungary. *Corvinus Journal of Sociology and Social Policy*, (2). Independent citations: 2

CURRICULUM VITAE

Senior Researcher

Surname: Elekes
First name: Zoltán
Date of birth: 08.08.1987.
ORCID: 0000-0002-7437-5791
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Current positions:

2021-: Researcher, Centre for Regional Science at Umeå University (CERUM), Umeå University, Sweden

2020-: Research Fellow, Agglomeration and Social Networks Lendület Research Group, HUN-REN Centre for Economic and Regional Studies, Budapest

Previous positions:

2019–2020: Senior research assistant, Umeå University, Department of Geography & CERUM, Sweden

2017–2020: Assistant research fellow, Hungarian Academy of Sciences, Centre for Economic and Regional Studies, Institute of Economics, Agglomeration and Social Networks Lendület Research Group, Hungary

2019–2020: Assistant professor, University of Szeged, Faculty of Economics and Business Administration, Institute of Economics and Economic Development, Hungary

2016–2018: PhD candidate, University of Szeged, Doctoral School of Economics, Hungary

2014–2019: Lecturer, University of Szeged, Faculty of Economics and Business Administration, Institute of Economics and Economic Development, Hungary

2014: Assistant research fellow, University of Szeged, Faculty of Economics and Business Administration, Institute of Economics and Economic Development, Hungary

2012–2015: PhD-student, University of Szeged, Doctoral School of Economics, Hungary

2012–2013: Research Assistant, University of Szeged, Faculty of Economics and Business Administration, Institute of Economics and Economic Development, Hungary

URL for web site: https://anet.krtk.mta.hu/researchers/z-elekes/

Education and key qualifications:

2018: PhD in Economics, University of Szeged, Doctoral School of Economics

2012: Economist, regional and environmental economics (MA), University of Szeged, Faculty of Economics and Business Administration, Hungary

Research experience:

2022-: principal investigator, FK OTKA grant No. 143064 project called "Structure and robustness of regional supplier networks".

2021-: principal investigator, FORTE grant No. 2020-00312 project called "Regions in change: Unpacking the economic resilience of Swedish labour market regions with respect to different groups of workers".

2021-: participating researcher, OTKA grant No. 138970 project called „Mobility, social- and economic networks, and inequality in cities during the pandemic”.

2019-2022: participating researcher, Swedish Research Council grant No 2017-02385 project called "When and where is it possible for young workers to escape from low-wage jobs? The role of the organizational and regional context for upward wage mobility."

2017–2022: participating researcher, Agglomeration and Social Networks Lendület Research Group, Hungarian Academy of Sciences, Centre for Economic and Regional Studies, Institute of Economics.

Received fellowships and grants:

2022-: FK OTKA grant No. 143064 project called "Structure and robustness of regional supplier networks". [18 M HUF]

2021-: FORTE grant No. 2020-00312 project called "Regions in change: Unpacking the economic resilience of Swedish labour market regions with respect to different groups of workers". [3.9 M SEK]

2021: Bárány Róbert Award, of the Eötvös Lóránd Research Network of Hungary, to recognize the work of outstanding young scholars across scientific disciplines.

2019: Excellent Young Regional Scientist Award of the Hungarian Regional Science Association, the Hungarian section of the European Regional Science Association.

2017–2018: New National Excellence Program of the Ministry of Human Capacities scholarship for PhD candidates to execute the research program called "The relationship between foreign-owned firms and agglomeration economies mediated by technological proximity."

2015: Campus Hungary Higher Education Staff Short Term Mobility grant for international conference participation.

The bibliographic data and the number of independent citations of the 5 most important publications over the last five years:

Elekes Z., Baranowska-Rataj, A., & Eriksson, R. (2023): Regional diversification and labour market upgrading: Local access to skill-related high-income jobs helps workers escaping low-wage employment. ***Cambridge Journal of Regions, Economy and Society***, rsad016. https://doi.org/10.1093/cjres/rsad016
Independent citations: 3.

Baranowska-Rataj, A., **Elekes Z.**, & Eriksson, R. (2023): Escaping from Low-Wage Employment: The Role of Co-worker Networks. ***Research in Social Stratification and Mobility***, 83: 100747. https://doi.org/10.1016/j.rssm.2022.100747
Independent citations: 3.

Tóth G., **Elekes Z.**, Whittle, A., Lee, C., & Kogler, D. F. (2022): Technology Network Structure Conditions the Economic Resilience of Regions. ***Economic Geography***, 98(4): 355-378. https://doi.org/10.1080/00130095.2022.2035715
Independent citations: 23.

Szakálné Kanó I., Lengyel B., **Elekes Z.**, & Lengyel I. (2019): Agglomeration, foreign firms and firm exit in regions under transition: the increasing importance of related variety in Hungary. ***European Planning Studies***, 27(11): 2099-2122. https://doi.org/10.1080/09654313.2019.1606897
Independent citations: 16.

Elekes Z., Boschma, R., & Lengyel B. (2019): Foreign-owned firms as agents of structural change in regions. ***Regional Studies***, 53(11): 1603-1613. https://doi.org/10.1080/00343404.2019.1596254
Independent citations: 71.

The bibliographic data and the number of independent citations of further 5 significant publications:

Elekes Z., & Lengyel B. (2020): Related trade variety, foreign-domestic spillovers and regional employment in Hungary. ***Acta Oeconomica***, 70(4): 551-570. https://doi.org/10.1556/032.2020.00036
Independent citations: 2.

Elekes Z., & Lengyel B. (2020): A külföldi tulajdonú vállalatok és az import szerepe a hazai térségek exportjának diverzifikációjában. ***Közgazdasági Szemle***, 67(4): 352-378. https://doi.org/10.18414/KSZ.2020.4.352
Independent citations: 1.

Tóth G., Juhász S., **Elekes Z.**, & Lengyel B. (2021): Repeated collaboration of inventors across European regions. ***European Planning Studies***, 29(12): 2252-2272. https://doi.org/10.1080/09654313.2021.1914555
Independent citations: 13.

Elekes Z., & Juhász S. (2017): A technológiai közelség által közvetített agglomerációs előnyök hatása a hazai vállalatok túlélésére. ***Tér és Társadalom***, 31(3): 3-24. https://doi.org/10.17649/TET.31.3.2873
Independent citations: 4.

Elekes Z. (2016): A regionális növekedés új tényezői az evolúciós gazdaságföldrajzi kutatásokban. A változatosság és a technológiai közelség. ***Közgazdasági Szemle***, 63(3): 307-329. http://dx.doi.org/10.18414/KSZ.2016.3.307
Independent citations: 13.

CURRICULUM VITAE

Participating Researcher

Surname: O.Szabo
First name: Rebeka
Date of birth: 10.12.1990.
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Current positions:

2024-: Assistant Professor at the Data Analytics and Information Systems Institute, Corvinus University of Budapest

2024-: Assistant Professor at Center for Collective Learning, Corvinus Institute for Advanced Studies, Corvinus University of Budapest

Previous positions:

2020-2024: Research Fellow at Laboratory for Networks, Technology & Innovations, Corvinus University of Budapest, Hungary.

2023: Teaching Assistant of "Introduction to Programming in Python" in MethodsNET summer school at Radbound University.

2022: Teaching Assistant of "Python 101" in winter school of GESIS – Leibniz Institute for Social Sciences.

2022: Instructor of "Python Programming for Social Sciences" in summer school of European Consortium for Political Research (ECPR).

2021, 2022: Teaching Assistant of "Python Programming for Social Sciences" in summer school of European Consortium for Political Research (ECPR).

2014-2015: Empirical Research Intern in Republikon Social and Political Research Institute, Budapest, Hungary.

URL for web site:

https://www.netilab.hu/researchers/rebeka-o-szabo/
https://networkdatascience.ceu.edu/people/rebeka-o-szabo

Education and key qualifications:

2022: PhD in Network and Data Science on the thesis "Problem-Solving Teams in Escape Rooms: Composition, Collaboration Networks, and Social Dynamics". Department of Network and Data Science, Central European University, Budapest, Hungary. Supervisor: Federico Battiston.

2016: MSc in Sociology – Comparative Organisations and Labour Studies. Faculty of Social and Behavioural Sciences, University of Amsterdam, The Netherlands

Research experience:

January 2020 – April 2020: 3 months stay as a visiting scholar at Kellogg School of Management and The Northwestern Institute of Complex Systems, Northwestern University, Chicago, IL, USA

February 2014 – August 2014: 6 months stay as an Erasmus student at the Faculty of Social Sciences, Vrije University, The Netherlands

Received fellowships and grants:
2022: Corvinus Research Excellence Award
2021: PhD Award for Advanced Students
2020: CEU Doctoral Research Support Grant
2018: CEU Research Support Scheme

The bibliographic data and the number of independent citations of the 5 most important publications over the last five years:

O. Szabo, R., Chowdhary, S., Deritei, D., Battiston, F. (2022). The anatomy of social dynamics in escape rooms. *Scientific Reports* 12. https://doi.org/10.1038/s41598-022-13929-0. Independent citations: 3

O. Szabó, R., Battiston, F., & Koltai, J. (2023). Faultlines, Familiarity, Communication: Predictors and Moderators of Team Success in Escape Rooms. *Small Group Research*, 0(0). https://doi.org/10.1177/10464964231183456 Independent citations: -.

Lőrincz L, Juhász S, **O. Szabó R**. (2023). Business transactions and ownership ties between firms. *Network Science*. 1-20. doi:10.1017/nws.2023.19. Independent citations idézők: -.

CURRICULUM VITAE

Postdoctoral Researcher

Surname: Toth
First name: Gergo
Date of birth: 14.08.1990
ORCID: 0000-0003-4461-0065
MTMT: 10073401

Current positions:

2023-: Postdoctoral Researcher at the Centre for Regional Science (CERUM), Umeå University, Umeå, Sweden

2018-: Junior Researcher at the Agglomeration and Social Networks Lendület Research Group (ANETI) at the Centre for Economics and Regional Studies, Hungarian Research Network, Budapest, Hungary

Previous positions:

2023: Lecturer at Rajk College for Advanced Studies on "Network Science".

2020: Lecturer at Széchenyi István College for Advanced Studies on "Network Science".

2020: External lecturer at the Department of Economics, Eötvös Loránd University on "Network Science".

2017-2018: Teaching Fellow at the Department of Economics, Eötvös Loránd University on "Introduction to Economics for Social Scientists", "Introduction to Economics for Computer Scientists" and "Thesis Writing Seminar for Economists".

URL for web site: https://gergototh.eu/

Education and key qualifications:

2023: PhD in Economic Geography on the thesis "Networks and Resilience: Essays on the Economic Structure of European Regions". Spatial Dynamics Lab, University College Dublin, Ireland. Supervisor: Dieter F. Kogler.

2017: MA in Political Science. Department of Political Science, Central European University, Budapest, Hungary

2014: BA in Applied Economics. Department of Economics, Eötvös Loránd University, Budapest, Hungary

Research experience:

2021-2022: Visiting researcher and research assistant at the Centre for Regional Science (CERUM), Umeå University, Umeå, Sweden

2016-2017: Research Assistant at the Institute of Advanced Studies, Central European University

2014-2015: Research Assistant at the Economics of Networks Research Group at the Centre for Economics and Regional Studies, Hungarian Academy of Sciences

Received fellowships and grants:

2023: Barany Robert Prize for outstanding research, Eötvös Loránd Research Network

2022: Junior Prima Prize in Hungarian Science category

2019-2023: Doctoral Scholarship, University College Dublin, Dublin, Ireland

2017-2018: Teaching Fellow Scholarship, Centre for Economic Research and Graduate Education, Economics Institute, Prague, Czechia

2015-2017: Master's Scholarship, Central European University

The bibliographic data and the number of independent citations of the 5 most important publications over the last five years:

Tóth G, Wachs J, Di Clemente R, Jakobi Á, Ságvári B, Kertész J, Lengyel B (2021) Inequality is rising where social network segregation interacts with urban topology. *Nature Communications* 12(1): 1143. DOI: 10.1038/s41467-021-21465-0.

Independent citations: 75.

Tóth, G., Elekes, Z., Whittle, A., Lee, C., & Kogler, D. F. (2022). Technology network structure conditions the economic resilience of regions. *Economic Geography*, 98(4), 355-378.

Independent citations: 23.

Tóth, G., & Lengyel, B. (2021). Inter-firm inventor mobility and the role of co-inventor networks in producing high-impact innovation. *The Journal of Technology Transfer*, 46, 117-137.

Independent citations: 19.

Juhász, S., **Tóth, G.**, & Lengyel, B. (2020). Brokering the core and the periphery: Creative success and collaboration networks in the film industry. *PloS one*, 15(2), e0229436.

Independent citations: 15.

Tóth, G., Juhász, S., Elekes, Z., & Lengyel, B. (2021). Repeated collaboration of inventors across European regions. *European Planning Studies,* 29(12), 2252-2272.

Independent citations: 13.

The bibliographic data and the number of independent citations of further 5 significant publications:

Gessler, T., **Tóth, G.**, & Wachs, J. (2022). No country for asylum seekers? How short-term exposure to refugees influences attitudes and voting behavior in Hungary. *Political Behavior*, 44(4), 1813-1841.
Independent citations: 39.

Czaller, L., **Tóth, G.**, & Lengyel, B. (2022). Allocating vaccines to remote and on-site workers in the tradable sector. *Scientific Reports*, 12(1), 4098.
Independent citations: -

Samu F., Czaller L., **Tóth G.**, Elekes Z., Lengyel B. (2022). A munkaerőpiac méretének szerepe a munkanélküliek preferált foglalkozásokban való elhelyezkedésében. In: Szabó-Morvai, Á; Lengyel, B (szerk.) *Munkaerőpiaci tükör, 2021: adminisztratív adatok a gyakorlatban*. Budapest, Magyarország: Közgazdaság- és Regionális Tudományi Kutatóközpont, 309 p. pp. 181-182., 2 p.
Independent citations: -

Tóth G., Lengyel B., Bíró A. (2022). A depressziót meghatározó szociális hálózati tényezők In: Szabó-Morvai, Á; Lengyel, B (szerk.) *Munkaerőpiaci tükör, 2021: adminisztratív adatok a gyakorlatban*. Budapest, Magyarország: Közgazdaság- és Regionális Tudományi Kutatóközpont, ELKH (2022) 309 p. pp. 167-170., 3 p.
Independent citations: -

CURRICULUM VITAE

Postdoctoral fellow

Surname: Gallo
First name: Luca
Date of birth: 04.28.1994.
ORCID: 0000-0002-2160-8467

Current positions:
2022-: Postdoctoral fellow at the Department of Network and Data Science, Central European University

Education and key qualifications:
2022: PhD in Complex Systems for Physical, Socio-economical and Life Sciences on the thesis "Coupled dynamical systems in presence of higher-order interactions". Doctoral School of Complex Systems for Physical, Socio-economical and Life Sciences, Department of Physics and Astronomy, University of Catania, Italy. Supervisors: Vito Latora, Mattia Frasca, Timoteo Carletti.
2019: MSc in Physics of Complex Systems. Department of Physics, University of Turin, Italy.

Research experience:
June 2021 – March 2022: 10 months visiting PhD student at the Namur Institute for Complex Systems, University of Namur, Belgium

The bibliographic data and the number of Google scholar citations of the 5 most important publications over the last five years:

Lotito QF, Contisciani M, De Bacco C, Di Gaetano L, **Gallo L**, Montresor A, Musciotto F, Ruggeri N, Battiston F (2023) Hypergraphx: a library for higher-order network analysis. *Journal of Complex Networks* 11(3), cnad09. DOI: 10.1093/comnet/cnad019. Google scholar citations: 10

Gallo L, Latora V, Pulvirenti A (2023) MultiplexSAGE: A multiplex embedding algorithm for inter-layer link prediction. *IEEE Transactions on Neural Networks and Learning Systems*. DOI: 10.1109/TNNLS.2023.3274565. Google scholar citations: -

Gallo L, Muolo R, Gambuzza LV, Latora V, Frasca M, Carletti T (2022) Synchronization induced by directed higher-order interactions. *Communications Physics* 5, 263. DOI: 10.1038/s42005-022-01040-9. Google scholar citations: 28

Gallo L, Frasca M, Latora V, Russo G (2022) Lack of practical identifiability may hamper reliable predictions in COVID-19 epidemic models. *Science Advances* 8, eabg5234. DOI: 10.1126/sciadv.abg5234. Google scholar citations: 19

Gambuzza LV, Di Patti F, **Gallo L**, Lepri S, Romance M, Criado R, Frasca M, Latora V, Boccaletti S (2021) Stability of synchronization in simplicial complexes. *Nature Communications* 12, 1255. DOI: 10.1038/s41467-021-21486-9. Google scholar citations: 155

The bibliographic data and the number of Google scholar citations of further 4 significant publications:

Muolo R, **Gallo L**, Latora V, Frasca M, Carletti T (2023) Turing patterns in systems with high-order interactions. *Chaos, Solitons & Fractals* 166, 112912. DOI: 10.1016/j.chaos.2022.112912. Google scholar citations: 23

Malizia F, **Gallo L**, Frasca M, Latora V, Russo G (2022) Individual-and pair-based models of epidemic spreading: master equations and analysis of their forecasting capabilities. *Physical Review Research* 4, 023145. DOI: 10.1103/PhysRevResearch.4.023145. Google scholar citations: 2

Gambuzza LV, Di Patti F, **Gallo L**, Lepri S, Romance M, Criado R, Frasca M, Latora V, Boccaletti S (2021) The Master Stability Function for synchronization in simplicial complexes. In*:* Battiston F, Petri G (eds) *Higher-Order Systems. Understanding Complex Systems. Springer, Cham.* DOI: 10.1007/978-3-030-91374-8_10. Google scholar citations: 1

Proverbio D, **Gallo L**, Passalacqua B, Destefanis M, Maggiora M, Pellegrino J (2020) Assessing the robustness of decentralized gathering: a multi-agent approach on micro-biological systems. *Swarm intelligence* 14, 313-331. DOI: 10.1007/s11721-020-00186-y. Google scholar citations: 1

CURRICULUM VITAE
Participating Researcher

Surname: Juhász
First name: Sándor
Date of birth: 09.07.1989.
ORCID: 0000-0003-3124-8597
MTMT: 10054212

Current positions:
2022-2024: Marie Skłodowska-Curie Postdoctoral Fellow, Complexity Science Hub Vienna

Previous positions:
2020-2022: Research Fellow at Laboratory of Networks, Technology and Innovation, Centre for Advanced Studies, Corvinus University of Budapest
2017-2020: Junior Research Fellow at Agglomeration and Social Networks Lendület Research Group, Hungarian Academy of Sciences, Budapest
2017-2019: Assistant Research Fellow, Intitute of Economics and Economic Development, Faculty of Economics, University of Szeged

URL for web site: https://sandorjuhasz.com

Education and key qualifications:
2020: PhD in Economics on the thesis “Foreign-owned firms, agglomeration externalities and knowledge flow”. University of Szeged, Hungary. Supervisor: Balázs Lengyel.
2019: PhD in Economic Geography on the thesis “Geography and Network Tie Formation”. Utrecht University, Netherlands. Supervisor: Ron Boschma.
2014: MA in Regional and Environmental Economic Studies. University of Szeged, Hungary
2011: BA in Business and Management. University of Szeged, Hungary

Received fellowships and grants:
2022: 199K EUR, from the European Union’s Marie Skłodowska-Curie Postdoctoral Fellowship Program “Supporting Local Economic Development with Business Transaction Data” (2022-2024)
2021: Corvinus Research Excellence Award
2018: ÚNKP National Excellence Award

The bibliographic data and the number of independent citations of the 5 most important publications over the last five years:

Juhász S, Broekel B, Boschma R (2021) Explaining the dynamics of relatedness: The role of co-location and complexity. *Papers in Regional Science* 100(1): 3-21. DOI: 10.1111/pirs.12567. Independent citations: 15.
Juhász S (2021) Spinoffs and tie formation in cluster knowledge networks. *Small Business Economics* 56(4): 1385-1404. DOI: 10.1007/s11187-019-00235-9. Independent citations: 18.

Bokányi E, **Juhász S**, Karsai M, Lengyel B (2021) Universal patterns of long-distance commuting and social assortativity in cities. *Scientific Reports* 20829: 10. DOI: 10.1038/s41598-021-00416-1. Independent citations: 10.

Juhász S, Pintér G, Kovács Á J, Borza E, Mónus G, Lőrincz L, Lengyel B (2023) Amenity complexity and urban locations of socio-economic mixing. *EPJ Data Science* 12(34): 1-18. DOI: 10.1140/epjds/s13688-023-00413-6. Independent citations: -

Tóth G, **Juhász S**, Elekes Z, Lengyel B (2021) Repeated collaboration of inventors across European regions. *European Planning Studies* 29(12): 2252-2272. DOI: 10.1080/09654313.2021.1914555. Independent citations: 13.

The bibliographic data and the number of independent citations of further 5 significant publications:

Juhász S, Lengyel B (2018) Creation and persistence of ties in cluster knowledge networks. *Journal of Economic Geography* 18(6): 1203-1226. DOI: 10.1093/jeg/lbx039. Independent citations: 33.

Juhász S, Tóth G, Lengyel B (2020) Brokering the core and the periphery: Creative success and collaboration networks in the film industry. *PLOS ONE* 15(2): e0229436. DOI: 10.1371/journal.pone.0229436. Independent citations: 15.

Kovács Á J, **Juhász S**, Boká**nyi** E, Lengyel B (2023) Income-related spatial concentration of individual social capital in cities. *Environment and Planning B* 50(4): 1072-1086. DOI: 10.1177/23998083221120663. Independent citations: -.

Lőrincz L, **Juhász S**, O. Szabó R (2024) Business transactions and ownership ties between firms. *Network Science* 1-20. DOI: 10.1017/nws.2023.19. Independent citations: -.

Juhász S, Lengyel B (2020) Kik formálják a klasztereket? Egy helyi tudáshálózat elemzése. *Területi Statisztika* 56(1): 46-65. DOI: 10.1371/ 10.15196/TS560104. Independent citations: 10.

Curriculum Vitae

Participating researcher

Surname: Bokányi
First name: Eszter
Date of birth: 20-10-1990
ORCID: 0000-0002-8625-1139
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Current position:
2021-: postdoc, POPNET project, University of Amsterdam

Previous positions:
2020 – 2021: postdoc, CIAS NeTI Lab, Corvinus University, Budapest
2019 – 2020: junior research fellow, HAS CERS ANET Lendület Group, Budapest
2017 – 2021: Mentor, Module Leader (Network Science) Milestone Institute, Budapest, Hungary
2016 – 2018: Teaching Assistant, Eötvös Loránd University, Budapest, Hungary
2015 – 2018: Data Analyst Intern (Smart Cities, Public Transport Optimization) Ericsson R&D Traffic Lab, Budapest, Hungary
2014 – 2015: Integration Course Teacher (Mathematics and Physics) Balassi Institute, Budapest, Hungary

URL for web site: https://www.illc.uva.nl/People/person/5210/Dr-Eszter-Bok%C3%A1nyi

Education and key qualifications:
2015 –2019: PhD in Physics, Eötvös Loránd University, Budapest, Hungary
2012 – 2015: Master in Physics, Eötvös Loránd University, Budapest
2014: Master in Physics, one semester program, Campus Hungary Scholarhip, Humboldt Universitat, Berlin
2009 – 2012: Bachelor in Physics, Eötvös Loránd University, Budapest
2009 – 2015: Mathematics and Physics Workshop, member, Eötvös József Collegium, Budapest

Received fellowships and grants:
2020: 2.4M HUF New National Excellence Program Postdoctoral Grant
2018: 3M HUF New National Excellence Program Doctoral Grant

The bibliographic data and the number of independent citations of the 5 most important publications over the last five years:

Kovács Á. J., Juhász, S., Bokányi, E., & Lengyel, B. (2022). Income-related spatial concentration of individual social capital in cities. Environment and Planning B: Urban Analytics and City Science, 0(0). https://doi.org/10.1177/23998083221120663

Bokányi, E., Novák, M., Jakobi, Á., & Lengyel, B. (2022). Urban hierarchy and spatial diffusion over the innovation life cycle. Royal Society Open Science, 9211038211038.

Bokányi, E., Juhász, S., Karsai, M., & Lengyel, B. (2021). Universal patterns of long-distance commuting and social assortativity in cities. Scientific Reports, 11(1), 1-10.
Number of independent citations: 3

Lengyel B., Bokányi, E., Di Clemente R., Kertész, J. & González, M. C. (2020). The role of geography in the complex diffusion of innovations. Scientific Reports 10, 15065.
Number of independent citations: 12

Bokányi, E. & Hannák, A. (2020). Understanding inequalities in ride-hailing services through simulations. Scientific Reports 10(1), 1-11.
Number of independent citations: 13

The bibliographic data and the number of independent citations of further 5 significant publications:

Bokányi, E., Kondor, D. & Vattay, G. (2019). Scaling in words on Twitter. Royal Society Open Science, 6(10), 190027.
Number of independent citations: 6

Bokányi, E., Kallus, Zs. & Gódor, I. (2019). Collective sensing of evolving urban structures: From activity- based to content-aware social monitoring. EPB: Urban Analytics and City Science, p. 239980831984876
Number of independent citations: 2

Bokányi, E., Szállási, Z., & Vattay, G. (2018). Universal scaling laws in metro area election results. PLOS ONE, 13(2), e0192913.
Number of independent citations: 11

Bokányi, E., Lábszki, Z., & Vattay, G. (2017). Prediction of employment and unemployment rates from Twitter daily rhythms in the US. EPJ Data Science, 6(1), 14.
Number of independent citations: 16

Bokányi, E., Kondor, D., Dobos, L., Sebők, T., Stéger, J., Csabai, I. & Vattay, G. (2016). Race, religion and the city: Twitter word frequency patterns reveal dominant demographic dimensions in the United States. Palgrave Communications. 2:16010
Number of independent citations: 23

CURRICULUM VITAE

Reseach Fellow

Surname: Pintér
First name: Gergő
Date of birth: 1990-01-07 (YYYY-MM-DD)
ORCID: 0000-0003-4731-3816
MTMT: 10054891

Current positions:

2022-: Research Fellow at Laboratory of Networks, Technology and Innovation, Centre for Advanced Studies, Corvinus University of Budapest

Previous positions:

2016-2022: University research assistant at SmartLab, Óbuda University

2015-2016: Software developer at Institute for Computer Science and Control, Hungarian Academy of Sciences (MTA SZTAKI)

URL for web site: https://pintergreg.github.io/

Education and key qualifications:

2022: PhD in Applied Informatics on the thesis “Analyzing the Mobility Customs of the Urban Population Using Mobile Network Data”. Doctoral School of Applied Informatics and Applied Mathematics, Óbuda University, Hungary. Supervisor: Imre Felde.

2015: MSc in Engineering Information Technology. John von Neumann Faculty of Informatics, Óbuda University, Hungary

The bibliographic data and the number of independent citations of the 5 most important publications over the last five years:

Sándor Juhász; **Gergő Pintér**; Ádám J. Kovács; Endre Borza; Gergely Mónus; László Lőrincz; Balázs Lengyel: *Amenity complexity and urban locations of socio-economic mixing*. EPJ Data Science volume 12, Article number: 34 (2023). DOI: 10.1140/epjds/s13688-023-00413-6

Gergő Pintér, Imre Felde: *Awakening City: Traces of the Circadian Rhythm within the Mobile Phone Network Data*. Information 2022, 13 (3); DOI: 10.3390/info13030114

Gergő Pintér, Imre Felde: *Commuting Analysis of the Budapest Metropolitan Area Using Mobile Network Data*. ISPRS International Journal of Geo-Information 2022, 11 (9); DOI: 10.3390/ijgi11090466, independent citations: 1

Gergő Pintér, Imre Felde: *Analyzing the Behavior and Financial Status of Soccer Fans from a Mobile Phone Network Perspective: Euro 2016, a Case Study*. Information 2021, 12 (11); DOI: 10.3390/info12110468, independent citations: 1

Gergő Pintér, Imre Felde: *Evaluating the Effect of the Financial Status to the Mobility Customs*. ISPRS International Journal of Geo-Information 2021, 10 (5); DOI: 10.3390/ijgi10050328, independent citations: 1

The bibliographic data and the number of independent citations of further 5 significant publications:

Gergő Pintér, Amir Mosavi, Imre Felde: *Artificial Intelligence for Modeling Real Estate Price Using Call Detail Records and Hybrid Machine Learning Approach*. Entropy 2020, 22 (12); DOI: 10.3390/e22121421, independent citations: 13

V. Institutional background of the planned research project

The Agglomeration and Social Networks Lendület Research Group (ANET Lab) has become a flagship group Centre for Economic- and Regional Studies (KRTK). First of all, spatial economic development and inequalities are core research problems addressed by the Institute of Economics and the Institute for Regional Studies of KRTK. Further, the members of the Lab have already contributed to further strategic aims of KRTK in many ways. First, the Lab increased the international embeddedness and impact of KRTK. Second, Lab members have collaborated intensively with other research groups advancing quantitative methodological approaches. Third, Lab members have contributed to the Centre's strategic aim to develop a new Big Data group in the Data Bank of KRTK. The recent proposal aims to continue these developments; thus, the management of KRTK provides full support to it.

KRTK offers outstanding capacities to contribute to the successful implementation of the project. It manages a Data Bank that coordinates the scientific use of Hungarian administrative datasets and provides server capacities for researchers to work with large datasets. The Data Bank maintains a Data Room at the Institute that is for working with sensitive data in a safe environment. The Data Bank has set up a new Big Data team that is responsible for the collection of online data and develop the data infrastructure to handle large datasets. A GDPR advisor helps the Institute to collect and work with sensitive data and surveys.

KRTK will allocate resources to the planned research group in the form of salaries and data access. The PI of the project is affiliated full time to KRTK and his salary adds up to 16.95 million HUF yearly including social benefits that is allocated by KRTK to the project. Two of the senior researchers are already affiliated part time and one is affiliated full time with KRTK, three of the postdoctoral researchers are already affiliated with KRTK. The base salary of these future colleagues that sum up to 32.49 million HUF yearly will be allocated to the project. Finally, KRTK has purchased mobility data in the amount of 25 million HUF plus taxes that will be accessible by the project free of charge. In addition to these, the Big Data group at the Data Bank of KRTK will allocate time to help data storage and collection occasionally (not quantified here). The contribution of KRTK to the project funds all add up to 272 million HUF over the 5 years.

The data collected in the project will be stored on the servers of the Data Bank of KRTK. The project will use their protocols to manage access for other KRTK researchers, external collaborators and visitors. The new data purchased within the project will be analyzed following the Data Bank regulations.

The reason for submitting this application for the Lendület H to be implemented in the current research entity is to ensure continuity of the Lendület I project that ended in 2022 and was evaluated "excellent". ANET Lab not only implemented the past research project successfully but became one of the most important Hungarian groups in regional science, network science, and computational social science. ANET Lab also had a big influence on the strategy of KRTK by facilitating data-driven research and had a direct impact on Corvinus University of Budapest as well in the form of a successful ERA-Chair application that aims institutional reform of data science research and education. By submitting the application to the Lendület H project, we aim to develop further the Budapest community of economists and social scientists doing research in social and spatial networks and also aim to strengthen the existing synergies with the widely defined network science community in Hungary. KRTK has proved itself to be an ideal location for these aims already.